Associated with document Ref. Ares(2022)3823781 - 20/05/2022

ANNEX 1

Horizon Europe (HORIZON)

Description of the action (DoA)

Part A

Part B

Associated with document Ref. Ares(2022)3823781 - 20/05/2022

DESCRIPTION OF THE ACTION (PART A)

COVER PAGE

Part A of the Description of the Action (DoA) must be completed directly on the Portal Grant Preparation screens.

PROJECT *Grant Preparation (General Information screen) — Enter the info.*	
Project number:	101058756
Project name:	REcycling technologies for Circular REuse and remanufacturing of fiber-reinforced composite mATErials
Project acronym:	RECREATE
Call:	HORIZON-CL4-2021-RESILIENCE-01
Topic:	HORIZON-CL4-2021-RESILIENCE-01-01
Type of action:	HORIZON-RIA
Service:	HADEA/B/03
Project starting date:	fixed date: 1 June 2022
Project duration:	48 months

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PROJECT SUMMARY

Project summary

Grant Preparation (General Information screen) — Provide an overall description of your project (including context and overall objectives, planned activities and main achievements, and expected results and impacts (on target groups, change procedures, capacities, innovation etc)). This summary should give readers a clear idea of what your project is about.

Use the project summary from your proposal.

The general objective of the RECREATE project is to develop a set of innovative technologies aimed at exploiting the potential of End-of-Life (EoL) complex composite waste as feedstock for profitable reuse of parts and materials in the manufacturing industry. In the light of this, more specific objectives are:
- to develop and validate in relevant environment (TRL6) novel reuse strategies for large EoL composite parts based on smart recognition and inspection for sorting (Laser Induced Breakdown Spectroscopy), high precision dismantling (laser-shock) and repair, temperature-assisted reshaping, design for disassembly based on reversible joints, AI-assisted decision support systems;
- to develop and validate in relevant environment (TRL6) innovative recycling technologies (catalyst-assisted green solvolysis and electrofragmentation) allowing simultaneous recovery of high quality, integer, clean fibers and of an organic resin fraction reusable as coating material, at the very end of the multiple reuse processes of parts;
- to demonstrate at TRL6 the use of smart and green reversible thermoset resins as enabling materials for the realization of the next generation of fiber-reinforced composites with easier repairability and enhanced reusability, facilitating the transition towards recyclable-by-design composites;
- to develop digital tools for the quantitative evaluation of the environmental and economic performance of the proposed technologies (LCA/LCC), as well as their circularity assessment and acceleration;
- to co-design innovative digital learning resources, based on MOOCs, gamification and digital twins of some specialty technologies developed in the project, with easy adoption and high replicability.
The objectives and ambition of RECREATE are fully compliant with the general requirements of the Horizon Europe - Digital, Industry and Space 2021 Work Programme, and with the specific requirements of the call Horizon-CL4-2021-Resilience-01-01.

LIST OF PARTICIPANTS

PARTICIPANTS

Grant Preparation (Beneficiaries screen) — Enter the info.

Number	Role	Short name	Legal name	Country	PIC
1	COO	POLIMI	POLITECNICO DI MILANO	IT	999879881
2	BEN	TAU	TAMPEREEN KORKEAKOULUSAATIO SR	FI	902999288
3	BEN	Fraunhofer	FRAUNHOFER GESELLSCHAFT ZUR FORDERUNG DER ANGEWANDTEN FORSCHUNG EV	DE	999984059
4	BEN	ICAM	ICAM OUEST	FR	998210996
5	BEN	UPAT	PANEPISTIMIO PATRON	EL	999894528
6	BEN	GAIKER	FUNDACION GAIKER	ES	999615071
7	BEN	CNRS	CENTRE NATIONAL DE LA RECHERCHE SCIENTIFIQUE CNRS	FR	999997930
8	BEN	INVENT	INVENT INNOVATIVE VERBUNDWERKSTOFFEREALISATION UND	DE	998801629

PARTICIPANTS

Grant Preparation (Beneficiaries screen) — Enter the info.

Number	Role	Short name	Legal name	Country	PIC
			VERMARKTUNG NEUERTECHNOLOGIEN GMBH		
9	BEN	IRIS	IRIS TECHNOLOGY SOLUTIONS, SOCIEDAD LIMITADA	ES	998993786
10	BEN	COBAT COMPOSITI	COBAT COMPOSITI	IT	889348963
11	BEN	RESCOLL	RESCOLL	FR	998135821
12	BEN	REST UG	REST UG	DE	896759957
13	BEN	BENASEDO	N. BENASEDO SPA	IT	984710439
14	BEN	Carbon Cleanup	CARBON CLEANUP GMBH	AT	892032953
15	BEN	G_MEDIA	GIACOMELLI MEDIA MANAGEMENT IN SVETOVANJE DOO	SI	914861709
16	BEN	GRIFO	GRIFO MULTIMEDIA SRL	IT	942558119
17	BEN	EDAG	EDAG ENGINEERING GMBH	DE	927231634
18	BEN	HEAD Sport GmbH	HEAD SPORT GMBH	AT	929476602
19	BEN	GEVEN SPA	GEVEN SPA	IT	986282033
20	BEN	APRA Europe	AUTOMOTIVE PARTS REMANUFACTURERS ASSOCIATION EUROPE	BE	933341858
21	BEN	AVK	AVK-INDUSTRIEVEREINIGUNG VERSTARKTEKUNSTSTOFFE EV	DE	998597735

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LIST OF WORK PACKAGES

Work packages

Grant Preparation (Work Packages screen) — Enter the info.

Work Package No	Work Package name	Lead Beneficiary	Effort (Person-Months)	Start Month	End Month	Deliverable No(s)
WP1	Input EoL streams and output requirements	10 - COBAT COMPOSITI		1	36	D1.1, D1.3, D1.2
WP2	Reuse and remanufacturing of EoL composite structures	3 - Fraunhofer		1	36	D2.4, D2.5, D2.3, D2.2, D2.1
WP3	Recycling and reprocessing of materials from non-reusable parts	1 - POLIMI		7	48	D3.4, D3.5, D3.3, D3.1, D3.2
WP4	Next-generation recyclable-by-design composites	8 - INVENT		10	36	D4.4, D4.2, D4.1, D4.3
WP5	Design and development of demonstrators	17 - EDAG		25	48	D5.1, D5.5, D5.7, D5.9, D5.11, D5.3, D5.4, D5.2, D5.6, D5.8, D5.10
WP6	Circular economy and sustainability assessment	5 - UPAT		13	48	D6.5, D6.4, D6.1, D6.2, D6.3
WP7	Co-design and validation of learning resources	16 - GRIFO		16	48	D7.2, D7.4, D7.6, D7.3, D7.5, D7.1
WP8	Communication, dissemination and exploitation	15 - G_MEDIA		1	48	D8.4, D8.1, D8.7, D8.8, D8.5, D8.3, D8.6, D8.2
WP9	Project management	1 - POLIMI		1	48	D9.2, D9.4, D9.3, D9.5, D9.1
WP10	Ethics requirements	1 - POLIMI		1	48	D10.1

Work package WP1 – Input EoL streams and output requirements

Work Package Number	WP1	Lead Beneficiary	10. COBAT COMPOSITI
Work Package Name	Input EoL streams and output requirements		
Start Month	1	End Month	36

Objectives
Map EoL composite waste source; describe the technical specifications of the recovered composite parts, fibers or resins to be used for demonstrators; define Key Performance Indicators (KPIs) for measuring and validating the RECREATE achievements; select EoL composite parts to be used both in the technical developments of WP2-4 and in the demonstrators (WP5). The outcomes of WP1 will put the basis for the development of technological solutions and will represent the starting point for the architecture definition of emerging digital business models (WP6).

Description
T1.1: Mapping of suitable EoL composite waste (leader: COBAT COMPOSITI; participants: all), M1-M6. A detailed assessment and mapping of the complex EoL composite waste stream sources in Europe in terms of composition, volumes, availability, and locations will be performed covering different sectors (wind, automotive, aviation, nautical, construction, etc). The main objective is to map waste streams to be used as input materials for the validation of the proposed reuse and recycling solutions (WP2,3), for the definition of suitable design guidelines for next-gen FRC materials and parts (WP4), and in the development of demonstrators (WP5). The output of this task will put the basis for emerging digital business models (WP6). T1.2: Definition of technical specifications for final demonstrators (leader: EDAG; participants: all), M1-M9. This task will focus on the definition of specific requirements for the reuse, recycling and remanufacturing of demonstrators. For each of the target products described in WP5, a schematic technical datasheet will be prepared, including (i) new product design, (ii) material properties, (iii) manufacturing process. Intermediate KPIs for the recovered composite parts, fibers or resins will be defined and checked during the development in WP2-4. Final target KPIs for measuring and validating the RECREATE achievements in the demonstrators (WP5) will be defined. T1.3: Selection and supply of EoL parts (leader: COBAT COMPOSITI; participants: IRIS, GAIKER, REST UG, GEVEN SPA, EDAG, BENASEDO, HEAD Sport GmbH, FRAUNHOFER), M2-M36. Based on the waste streams analyzed in T1.1 and technical specifications of T1.2, in this task the selection and the supply of EoL composite parts to be used both in the technical developments of WP2-4 and in the demonstrators (WP5) will be formalized. A schematic form will be proposed to the technological partner in order to collect information about type and quantity of required materials. According to the specific request, different EoL wind blades will be provided through the Cobat Compositi network.

Work package WP2 – Reuse and remanufacturing of EoL composite structures

Work Package Number	WP2	Lead Beneficiary	3. Fraunhofer
Work Package Name	Reuse and remanufacturing of EoL composite structures		
Start Month	1	End Month	36

Objectives
Develop effective strategies for reuse and remanufacturing of whole EoL composite complex structures, allowing a true closed-loop circular economy of composite parts. Advanced methods will be developed for material identification and automatic evaluation of such complex structures (i.e., LIBS, AI, NDT, image processing, XEIDANA, software development), which will serve as efficient decision-support tools for part reuse and remanufacturing. In addition, suitable inspection, testing, defect characterization and repair protocols will be defined and implemented to ensure effective part recovery and reusability. Different strategies for direct reuse and remanufacturing of complex composite components will be considered, including laser-shock-assisted active dismantling of multilayer structures (for de-coating,

disassembly and adhesive disconnection) as well as T-assisted curved-parts re-shaping. Results will be transferred to demo-case owners for further development/validation (TRL6).

Description
T2.1: Smart recognition and monitoring techniques for sorting (leader: IRIS; participants: FRAUNHOFER, CNRS, GAIKER), M1-27. This task will focus on the identification of FRCs with respect to fiber/matrix composition (e.g., polyester-based GFRCs, epoxy-based CFRCs). The feasibility of laser-induced breakdown spectroscopy (LIBS, alone and/or in combination with other suitable photonic technologies) will be assessed. Materials coming from preselected flows (T1.1/1.3) together with their composition characterization and others ad-hoc samples supplied by other partners (e.g., GAIKER) will be collected and supplied to T2.1. IRIS will assess the accuracy in the materials identification by the photonic technology to select the most appropriate configuration, and will develop preliminary AI hybrid models based on chemometric and machine learning principles. Complementary identification activities will also be carried out by means of dedicated image processing and evaluation tools developed at FRAUNHOFER. T2.2: Non-destructive inspection, defect characterization and repair (leader: FRAUNHOFER; participants: INVENT, ICAM, CNRS, RESCOLL), M1-36. In this task, different non-destructive approaches for inspecting EoL composite parts will be implemented to enable efficient defect detection, characterization and repair. Among these, active and passive thermography, X-ray CT, ultrasound-based and radar-based technologies will be used for inspection during production, in operation as well as during and after repair. Decision on the fate of the EoL FRC components will be made possible regarding the potential for part reuse or recycling. Component scanning and image data processing will be used for reverse engineering to develop the fitting/repair design strategy by capturing the real image contour of the EoL part, so that the repair structure can be manufactured to fit (first time right). Suitable methods (acoustic emission, adapted mechanical testing or others) will be implemented on the prototype line. T2.3: Decision Support System (DSS) for part re-use (leader: POLIMI; participants: FRAUNHOFER, CNRS, RESCOLL), M7-30. The output of non-destructive testing activities (T2.2) including e.g. images from active thermography and ultrasound will need to be evaluated automatically for improved process throughput. For this purpose, dedicated software tools available at FRAUNHOFER (e.g., XEIDANA) will be further developed. A dedicated Decision Support System (DSS) will be developed, able to elaborate the information acquired and the detailed end-use requirements for the different components and assemblies, fed by the activities carried out in T1.1 with the support of all democase leaders (WP5). T2.4: Active dismantling of complex and multilayer structures (Leader: CNRS; participants: FRAUNHOFER, RESCOLL, IRIS), M10-36. A novel efficient and selective dismantling process using laser wave shock technology with double and symmetrical impact configurations will be developed by CNRS with the support of RESCOLL team and facilities. Physical/digital twins based on FEM or/and reduced model will be developed and implemented. Efforts will be undertaken to optimize the dismantling process so as to ensure integrity of the dismantled material. On-line process control using active/passive sensors will be implemented. Materials will be inspected by NDT (ultrasonic) techniques (T2.2). The targeted development will start from preliminary testing on in-house manufactured GFRC/CFRC laminate samples (TRL3) and will be ultimately directed towards active dismantling of representative (T1.3) EoL composite parts (TRL6). Efficient methods will also be developed (FRAUNHOFER) for complete removal of paints and coatings on EoL FRC structures to enable effective recycling (WP3) and reuse for remanufacturing. T2.5: Temperature-assisted reshaping (leader: ICAM; participants: TAU, POLIMI, GAIKER), M10-36. A novel cost-effective process to directly reuse large curved EoL parts made out of polyester/vinylester GFRCs will be developed. The new technology development currently demonstrated at TRL3 by ICAM will be based on reshaping large GFRC parts with simple geometry using pressure and temperature at the maximum deformation point of the impregnating polyester resin (i.e., at the glass transition temperature) without polymer degradation. A systematic investigation will be done of the stress relaxation behaviour of materials followed by the optimization of relevant process parameters of the reshaping step. This process will be brought up to TRL6 via two synergistic strategies: batch process (GAIKER) using a suitably modified compression molding machine; semi-continuous process (ICAM) using an adapted twin-roll calendering machine. The experimental results will be then transferred to WP5 to be further validated at TRL6. A further advancement of this approach will be the synergistic development of the reshaping technology along with the study of new resin systems for the future generation of FRC materials (WP4).

Work package WP3 – Recycling and reprocessing of materials from non-reusable parts

Work Package Number	WP3	**Lead Beneficiary**	1. POLIMI
Work Package Name	Recycling and reprocessing of materials from non-reusable parts		
Start Month	7	**End Month**	48

Objectives
Develop (up to TRL6) innovative composite recycling technologies able to produce liberated fibers with minimal damage and loss of physical properties. Specific tasks will be dedicated to electrofragmentation and catalyst-assisted low-T green solvolysis, the latter technology being able to generate not only high quality fibers but also a reusable organic fraction. Other tasks will involve the recycled fiber and composite material characterization, and the definition of experimental protocols for fiber resizing and composite reprocessing. Finally, a techno-economical validation of the new technologies proposed will be done benchmarking the obtained characterization results against standard values available in the market, e.g. pyrolysis for CFs and mechanical grinding for GFs.

Description
T3.1: Electrofragmentation (leader: POLIMI; participants: TAU), M7-36. The goal of this task is the development of an efficient size-reduction process based on high-voltage fragmentation (HVF). HVF is an innovative technology for size-reduction of multiphase and/or multilayer materials which exploits a fast-growing voltage to overcome the dielectric breakdown threshold of the product under process. As HVF is only available at lab scale for recycling (TRL3), this task will perform first of all an extensive experimental campaign to fully understand the influence of the different parameters on the output material. The obtained results will be exploited for the creation of a cyberphysical model of the process itself, based on data acquisition and digital simulation, leading to process optimization, enabling its scale-up to TRL6. Clean fibers as well as matrix residues will be characterized in T3.3. T3.2: Catalyst-assisted low-temperature green solvolysis (leader: POLIMI; participants: TAU, ICAM), M7-36. A mild solvolysis process will be developed based on the selection of sustainable solvents (atoxic, and preferably biobased). Suitable catalyst to assist depolymerization will be used; this approach will allow to carry out the process at moderate temperature and ordinary pressures, minimizing energy consumption and maximizing the quality of the fibers recovered. Different catalyst will be tested, like for example TBD (triazabicyclodecene) for polyester hydrolysis. Experimental campaigns will involve both currently available but not reusable EoL composite waste, as well as future-generation polymer composites based on reversible resins or vitrimers. Characterization will include thermogravimetric analysis and electron microscopy to assess the purity of the recovered fibers, and chromatographic and spectroscopic methods (HPLC, GPC, NMR) to evaluate the organic liquid fraction, in view of its potential use as coating material (see Task 5.9). The optimized process will be scaled up for the supply of recycled fibers and resins for the demonstrators (WP5). T3.3: Materials characterization, preparation and reprocessing (leader: TAU; participants: POLIMI, ICAM, FRAUNHOFER, Carbon Cleanup), M10-48. A refined characterization of short individual recycled fibers from T3.1 and T3.2 will be performed. The characterization includes analysis of the fiber surface properties with microscopical and spectroscopical methods, and the evaluation of mechanical properties with micro-robotic test system. In this task, also activities related to the re-sizing of recycled fibers provided by T3.1/T3.2 will be undertaken to ensure the required static and dynamic properties in second-generation GFRC/CFRC parts (WP4). To this end, both wet chemistry and vapor-phase methods will be applied to maximize recycled fibers adhesion with the matrix in second generation FRCs. The effect of re-sizing on the composite properties will be analyzed in micro-, meso and macro-scales with a micro-robotic device as well as with electron microscopy analysis. The re-sizing will be integrated to the re-alignment of the recycled fibres in collaboration with ICAM and Carbon Cleanup. In particular, rCFs obtained in discontinuous format will be reprocessed into mats via an upgraded carding and continuous roving method based on fiber alignment and binding (Carbon Cleanup), ultimately enabling the manufacture of fabrics, mats, and ultimately prepregs based on recycled fibers to be used in the demo-cases developed in WP5. T3.4: Benchmarking and process validation of new recycling technologies (leader: Carbon Cleanup; participants: COBAT COMPOSITI, POLIMI, ICAM, FRAUNHOFER, TAU, HEAD Sport GmbH), M28-48. Based on the SoA and ongoing development of the recycling technology available at Carbon Cleanup and the validation of this process, a broader benchmarking activity will be carried out in this task to qualify and validate the recycling

technologies developed in this project (WP3) with currently used homologues. In particular, selected SoA recycling processes such as pyrolysis and mechanical shredding will be used as reference technologies to validate the approaches proposed in RECREATE. Key validation parameters will be focused on sustainability (e.g., CO2 footprint, energy consumption, waste generation), process (output material quality, throughput – in collaboration with T1.3) as well as on economic (profitability, scalability) factors.

Work package WP4 – Next-generation recyclable-by-design composites

Work Package Number	WP4	**Lead Beneficiary**	8. INVENT
Work Package Name	Next-generation recyclable-by-design composites		
Start Month	10	**End Month**	36

Objectives
Develop technologies to enhance the reusability and recyclability of composite components. The main route for this activity is the development of reversible joining technologies, based on modified conventional and green/smart resins. Design rules will be developed and provided for modular composite parts with reversible joints.

Description
T4.1: Modular design of next-gen composite structures (leader: INVENT; participants: FRAUNHOFER, EDAG), M10-30. Aim of this task is the development of a modular design methodology for composite parts that allows to separate the products in functional items to enable reusability of components and structures. Approaches for a modular design will be derived from the use cases (wind rotor blades, aircraft interior, automotive parts etc.). Aspects of efficient dismantling, suitable interfaces, main directions of stress and further aspects will be considered for the design of the modular next-gen composite items. Generalizations will be made from the use cases in order to finally establish generally applicable design rules. INVENT together with the dedicated partners will analyze the different parts from the use-cases to define the best possible modularity. Redesigns will be performed together with the use-case owners to verify and demonstrate the modular design (virtually and/or by means of generic or full demonstrators). FRAUNHOFER will contribute to this activity by classification of FRP components or products into recycling and reuse structures. T4.2: Design and requirements for reversible joining (leader: EDAG; participants: FRAUNHOFER, RESCOLL), M13-33. Aim of the task is to develop guidelines and calculation sheets for reversible adhesive joints taking into account the properties of the adhesive, the processes and the different scenarios for end uses. Specific sets of requirements will be developed in this task for the industrial sectors addressed in this project through the demo-cases described in WP5. This covers the production, the utilization, the dismantling and the reproduction of next-gen FRC parts and products. These sets of requirements will be used to target the design of materials and related remanufacturing processes in the context of a new generation of circular, easy to disassemble and reuse FRCs. In addition, a classification of existing joint connections and material combinations with regard to reversibility will also be provided (FRAUNHOFER), enabling the definition of a design methodology for reversible joints. T4.3: Reversible adhesives and joints (leader: RESCOLL; participants: FRAUNHOFER, EDAG), M16-36. In this task, RESCOLL will develop dedicated debondable adhesives and primers based on specific binders (epoxy, polyamides, …) and blowing agents to allow the easy separation of dissimilar assemblies (i.e., composite panels on metal structure, composite/composite assemblies), where monomaterial strategies cannot be implemented. The solutions provided by RESCOLL will be heat triggered (100-200 °C) with non-reversible weakening of the joints for easy manual dismantling and post-cleaning. RESCOLL will first conduct works at coupon level (TRL3/4) and produce large parts on demonstrator level, with automated deposition systems (TRL6). EDAG will support RESCOLL in the definition of the specs of the bonds and the supply of the coupons and the demonstrator parts (WP5). T4.4: Reversible green resins for multiple reuse, repair and recycling (leader: POLIMI; participants: RESCOLL, BENASEDO), M16-36. The main focus of this task will be on the development of sustainable (styrene and formaldehyde free) resins for next generation FRCs based on bioderived and/or “smart” reversible vitrimers and Diels-Alder (DA) compounds. These systems are featured by reversible crosslinking chemistry which enables their easier temperature-assisted reuse,

repair and recycling in next-gen FRCs. Thermally-reversible epoxy-based thermosets will be developed exploiting the mild temperature (50 °C) D-A cycloaddition reaction between selected monomers bearing furan groups (e.g., multifunctional glycidyl ether-based furfuryl derivatives) and maleimide moieties. In parallel, bioderived vitrimer epoxy matrices will be developed with characteristic target temperatures (Tg and Tv) compliant with the requirements of the ultimate demo-cases application (WP5). Metal-based catalytic systems will be considered (e.g., zinc acetate, triphenyl phosphine, triazabicyclodecene) for the dynamic interchange reactions. Process parameters such as catalyst loading and monomer stoichiometry will be carefully controlled to optimize kinetics of the exchange reactions (viz., their thermal processability) as well as crosslinking density (viz., their thermal stability).

Work package WP5 – Design and development of demonstrators

Work Package Number	WP5	**Lead Beneficiary**	17. EDAG
Work Package Name	Design and development of demonstrators		
Start Month	25	**End Month**	48

Objectives
Develop at pilot-level (TRL6) demonstrators of new products combining the reuse and recycle approaches of GFRCs/CFRCs developed in WP2-4. Different potential applications for reused and recycled GFRCs/CFRCs in real-case products covering a diverse array of market sectors will be demonstrated in industrially relevant environment. The overall functional validation will also be performed in accordance with the requirements outlined in WP1.

Description
T5.1: Demo-case reuse of EoL GFRC parts in technical boards for transportation (leader: GAIKER; participants: ICAM), M25-45. EoL PEs-based GFRC flat panels will be used to develop technical boards for the transport sector validating the reshaping technology developed in T2.5. The performance of reused/remanufactured parts will be evaluated so as to comply with the following technical KPIs: bending strength UNE EN 310 30-40 MPa; modulus of elasticity in bending UNE EN 310:> 7.000 MPa; water absorption: <3% UNE EN ISO 62; thermal conductivity <0,033 w/m ºC (UNE EN 12667); acoustic isolation > 28 db ASTM E1050-98 and ISO 10534-2:1998; weight < 12kg/m2 with thickness between 6-12mm; thermal stability between -20 to 80ºC; good abrasion resistance and scratch resistance. A minimum dimension coupon of 500x800mm2 will be addressed. Depending on the properties of the final reshaped panel (results from T2.5), two different strategies will be followed: for reshaped panels with thickness > 5mm and bending strength > 30MPa, a panel with appropriate finish coating will be developed; for reshaped panels with thickness < 5mm and bending strength < 30MPa, a sandwich structure will be developed, and the reshaped panel will be used as external cover of a polymeric core or of a sandwich structure. T5.2: Demo-case reuse of EoL CFRC parts in modular structures for mobility (leader: EDAG; participants: RESCOLL), M25-45. The aim of this task is to create a novel body-in-white concept using CFRC profiles and connection elements. This body-in-white will be developed for a vehicle of the future, e.g. a modular transportation system for people and goods. The profiles will be dismantable, remanufaturable and by this reusable. It is planned to achieve a set of standard elements that can be produced easily and connected via standard connectors to achieve interchangeability. Specifically, the activities in this task will take advantage and use of the novel technologies for dismantling developed in WP2 and will anticipate these technologies with a design for reuse based on the requirements set out in T1.2. The demonstrator will also include the simulation of e.g. stiffness and all performance-related properties. The design will be modular to enable changes in the concept that will be necessary if further reuse of the components is planned. (Target KPIs: >80% crash performance compared to classic design with new materials). T5.3: Demo-case reuse of EoL CFRC parts in segmentable rotor blades for wind energy (leader: REST UG; participants: INVENT, FRAUNHOFER, RESCOLL), M25-45. In this task, a segmentable rotor blade for a wind turbine will be demonstrated. Based on the results of T4.2 (Concept and Design) and T4.3 (Reversible Adhesives and Interconnects), a suitable physical demonstrator will be produced to demonstrate the modularity of the rotor blade and the reusability of structures and components. This requires the conceptual design and development of suitable manufacturing technologies. In addition, the ease of assembly and disassembly of the rotor blade segments will be assessed and validated. The physical demonstrator will be supplemented

by various simulations of a complete segmentable rotor blade in a realistic application environment to prove that all requirements (static and dynamic loads, aerodynamics) are met. Remanufacturing and repair methods for the surface and core structure of segmented rotor blade as the result of T2.3 will be demonstrated in trails using the physical rotor blade demonstrator. Innovative process technologies such as mobile robots and various NDT methods (active thermography, ultrasound, scanning – T2.2) will be used to assess component quality both after EoL and after repair and remanufacturing and to determine material flow. (Target KPIs: ≥ 50% substitution of virgin fibers; 30% CO2 emission reduction; 20% cheaper/faster repair process vs. ref. blades).

T5.4: Demo-case reuse of EoL GFRC/CFRC parts for aircraft cabin interiors (leader: GEVEN SPA; participants: FRAUNHOFER, RESCOLL), M25-45.
In this task, GEVEN SPA will focus on panels for aircraft seat components and cabin interiors for passenger Cabin and Service Area. The first goal will be to replace currently used seat backrests involving the use of aluminum alloys with reused/reshaped epoxy-based CFRCs obtained in the form of flat or slightly curved panels. The second goal will be to replace the currently used aircraft cabin panels based on Nomex-cored sandwich architecture coupled with CF reinforced skins, with reused/reshaped epoxy-based GFRC sandwich panels. For both demonstrators, GEVEN SPA will exploit its 'burner facility' and will perform dedicated experimental tests at coupon level vs. flammability (CS 25.856 Appendix F, Part I) and smoke/toxicity, so as to integrate them in the major items achieving TRL4. In addition, prototypes of panels for a passenger backrest seat and cabin interiors will be developed and validated up to TRL6 by means of dedicated experimental tests for the resistance to static, dynamic and "abuse loads", as well the aesthetic harmonization with cabin environment through the application of decorative, protective and reflective films. (Target KPIs: 8%÷12% weight-saving vs. market-available standard; 5%÷10% RC cost-saving vs. reference process; 12%÷25% fracture toughness increase vs. reference CFRCs).

T5.5: Demo-case reprocessing of recycled GFs/CFs and next-gen resins for aircraft cabin interiors (leader: GEVEN SPA; participants: TAU, REST UG), M25-45.
GEVEN SPA will focus on Cabin Lining Panels for passenger Cabin and Service Area. The main goal will be to design more cost-effective sidewall window panels using newly developed resins (T4.4) reinforced with rGFs/rCFs (WP3). For such demonstrator, GEVEN SPA will exploit its 'burner facility' and will perform dedicated experimental tests at coupon level vs. flammability (CS 25.856 Appendix F, Part I) and smoke/toxicity, so as to integrate them in the major items achieving TRL4. In addition, prototypes of panels for cabin lining panels will be developed and validated up to TRL6 by means of dedicated experimental tests as in T5.4. (Target KPIs: 8%÷12% weight-saving vs. market-available standard; 5%÷10% RC cost-saving vs. reference process).

T5.6: Demo-case validation of smart recognition and monitoring technologies for sorting (leader: IRIS; participants: all), M25-45.
IRIS will develop and build a recognition monitoring prototype unit bringing together laser-induced breakdown spectroscopy (LIBS, alone and/or in combination with other feasible photonic technologies) defined in Task 2.1, engineering aspects (e.g., conveyor, samples positioning) and the advanced AI and chemometric techniques embedded in a customized software developed for this purpose. Finally, Factory Acceptance Tests (FAT) will be performed to guarantee that all the components are running properly and the monitoring system can operate in an industrial environment. The validation at TRL6 will be carried out in IRIS facilities through Site Acceptance Tests (SAT). With the support of Consortium partners, especially providing pre-selected batches of materials with different known compositions, which will be monitored by the system, the results of the materials identification will be provided as validation of the system.

T5.7: Demo-case reprocessing of recycled CFs and next-gen resins for sport equipment (leader: HEAD Sport GmbH; participants: Carbon Cleanup, POLIMI, TAU), M25-45.
HEAD Sport GmbH will use rCFs from solvolysis and/or HVF to produce parts of rackets with a flat surface (typically, padel/pickle-ball rackets) in the form of sheet molded compounds. Both standard impregnating resins as well as newly developed biobased and smart resins (T4.4) will be used and benchmarked through different demonstrators. For this demonstrator, HEAD Sport GmbH will use its facilities in Kennelbach for testing at coupon level and to produce prototypes. Demonstrators will be tested according to the relevant standards and sporting conditions. (Target KPIs: processability in standard lines, fatigue cycle of 10.000 ball impacts).

T5.8: Demo-case reprocessing of recycled GFs for sport equipment (leader: HEAD Sport GmbH; participants: POLIMI), M25-45.
HEAD Sport GmbH will use rGFs (from WP4) as replacement for virgin GFs to produce a rGF-based fleece to be further impregnated and incorporated in the structural architecture of amateur skis. For this demonstrator, HEAD Sport GmbH will use its facilities in Kennelbach for testing at coupon level and to produce prototypes. Demonstrators will be tested

according to the relevant standards and sporting conditions. (Target KPIs: processability of fleece on prepreg line, resin uptake >100%, delamination test > 0.8 N/mm).

T5.9: Validation of recycled organic fraction for industrial coatings (leader: BENASEDO; participants: POLIMI, TAU), M25-45.
The organic fraction recovered by solvolytic experiments in various conditions (T3.2 and T4.4) will be evaluated as base component for coating formulations. Depending on the nature of the solvent and resin-type (polyesters or epoxies, partially depolymerized), the fraction will be suitably repolymerized or directly mixed with different binders (e.g., alkyd resins). The final product will be applied on metal substrates and will be evaluated in terms of film forming properties, hardness, adhesion and corrosion resistance with the goal to develop anticorrosion primers or base-coats. The demonstrator will be delivered in the form of painted metal parts. (KPI: dimensions 500 mm x 500 mm; pencil hardness: > B; adhesion: cross-cut test 100%; corrosion resistance: 1000 h salt-fog test).

T5.10: Monitoring and assessment of demonstrators against KPIs (leader: POLIMI; participants: all demo-case leaders), M31-48.
This task, an intermediate and an ex-post analysis of the demonstrators based on the KPIs identified in T1.2 will be performed, through a quantitative evaluation of the technical performance of the new products defined in T5.1-5.9 in relevant environments. In the intermediate analysis, the activities will be focused on the assessment of the status of the demonstrators, on the evaluation of the specific approach designed for integration of the solutions developed in WP2, 3 and 4 in the demonstrators, and on the update of the target KPIs considering the validation and testing of the developed solutions.

Work package WP6 – Circular economy and sustainability assessment

Work Package Number	WP6	**Lead Beneficiary**	5. UPAT
Work Package Name	Circular economy and sustainability assessment		
Start Month	13	**End Month**	48

Objectives
Develop a general circular business plan for reused and recycled composite parts and materials; implement the circular economy of composite parts and materials with dedicated digital technologies; implement an integrated Environmental and Socio-Economic Sustainability Assessment of reused and recycled composite parts; evaluate and rank the available reuse and recycling technologies of FRC materials developed in WP2-4.

Description
T6.1: Innovation management and circular business plans (leader: G_MEDIA; participants: UPAT, EDAG), M07-M36. A comprehensive innovation management programme will be developed by G_MEDIA. First, technology concepts will be verified in a controlled environment designed in innovation workshops organised to engage all participating actors for each of the streams by construing upon hypothetical scenarios and design testing pathways. Second, component and breadboard validation in relevant environment will be carried out for each tested technology. In step three, a feedback look will be used to improve the circularity potential of each proposed technology concept and to identify key competencies to be fostered in research and development teams for each concept. A circularity assessment tool will be developed by G_MEDIA and applied at the beginning of the programme and at the entry into each cycle of the three stages of the innovation programme, as well as at its completion. EDAG (WP5 leader) will participate to the workshops as champion partner to define and test the newly established business models by creating novel products for the mobility industry. T6.2: Digital tools for the demonstration and acceleration of circularity (leader: UPAT; participants: EDAG, all), M13-M48. UPAT will develop dedicated digital tools such as the Internet of Things (IoT), big data and data analytics that will be used by the demo-case leaders/end-users to accelerate circular technologies. Digital technologies will enable the circular economy in six different ways: overcoming logistical roadblocks around raw material collection for reuse and recycling; sorting high-quality feedstock and tracking the quality of the EoL and recycled materials; separating and recycling mixed/complex-material products; enhancing the yield and stability of the reuse and recycling processes; designing composite parts for reuse and recycling; using emerging digital business models to enable reuse and recycling. UPAT will apply

the digital tools to at least 3 demonstrators proposed in WP5 to evaluate the degree of enhancement of circularity and will deliver digital platforms to be used by the end-users.

T6.3: Integrated environmental and socio-economic sustainability assessment (leader: UPAT; participants: EDAG, all), M13-M48.
In this task, an integrated environmental and socio-economic sustainability assessment (ESESA) will be performed by UPAT to address the environmental, economic and social benefits from the reuse, recycling and remanufacturing processes. The framework of the ESESA is derived from the ISO 14040/14044 and inspired by the socio-LCA. The evaluations will be mainly performed using commercial software tools and (wherever necessary) customized modules will be developed by UPAT. EDAG, as champion demo-case and WP5 leader, will provide data and participate in optimizing the carbon footprint of the targeted mobility products. The products will generate novel utilization models and influence the collaboration of business partners and customers and by this also generate other influences on the environment. The business models of the products will be examined to identify profitable businesses in the circular supply chain. In the EoL analysis, a specific index will be used as a decision support tool for choosing the optimal reuse/recycling process among the alternative technologies proposed for the waste of the demonstrators. This index is a function of the quality of the selected material, the cost of the selected process and the environmental footprint of the process.

Work package WP7 – Co-design and validation of learning resources

Work Package Number	WP7	**Lead Beneficiary**	16. GRIFO
Work Package Name	Co-design and validation of learning resources		
Start Month	16	**End Month**	48

Objectives

Co-design, develop and validate new e-learning resources related to the broad interdisciplinary field of composite circularity. The learning resources will aim at transferring to employees and students the core skills and competencies needed for the uptake of the resilience solutions developed in the project, according to a cross-subject approach. The main objectives of this WP are: identify the most appropriate teaching methodologies and strategies to achieve objective-oriented learning for the intended target audience; define the digital tools to be used to implement the learning resources; co-design the learning resources together with subject matter experts (academic and RTO partners in RECREATE), representatives of regional institutional bodies, other external academic partners, local educational organizations; practically implement the learning resources; validate them in pilot testing sessions within the learning offering of the MOOC platform available at POLIMI.

Description

T7.1: Co-design of innovative learning/teaching methods (leader: POLIMI; participants: GRIFO, AVK), M16-M42.
In this task, the METID unit @POLIMI will co-design a set of Teaching Methodologies and Strategies Guidelines for RECREATE with the involvement of relevant stakeholders (representatives of RTO and universities, local authorities like sectorial associations). Starting from an in-depth analysis of the the specific needs of the different categories of learners identified (current and future professionals, undergraduates), this document will define the active and problem-driven approach to learning that we intend to apply, taking into account also the requirements of modularity and flexibility of the components of the learning resources. Such Guidelines will include the description of a simulation learning system, capable of modeling the process of interest and allowing the students to interact with a more manageable model of reality. The learning materials will be provided in input from WP2, 3, 4 and 6. Different pedagogical approaches (i.e. scenario-based learning, problem-based learning, project-based learning, challenge-based learning, inquiry-based learning, Kolb's cycle etc.) will be proposed. Within this framework, POLIMI will define all the necessary Teaching Methodologies and Strategies Guidelines to create a digital learning path in line with the project needs and objectives.

T7.2: Technological implementation of learning resources within a game environment (leader: GRIFO; participants: POLIMI), M19-M45.
In this task GRIFO will technically develop the learning resources defined in the Guidelines and the storyboards delivered as output in T7.1. The learning resources will be delivered through a gamification platform that will embed them in a specific storytelling. The use of gamification tools will facilitate the understanding of the key aspects of the subject matters, and support the engagement of learners. GRIFO will develop a specific instance of its proprietary gamification platform, which allows to develop and deliver a training course based on gamification dynamics that can be defined

flexibly based on the specific objectives to be achieved. Each action performed by the learner within the platform is associated with points, badges and general rewards. Early in the development phase, GRIFO will identify a specific theme for the visual map that will illustrate the learning path. The learning path will include a simulation system, to be integrated at a specific point in the process.

T7.3: Digital twins of key technologies (leader: GRIFO; participants: POLIMI, FRAUNHOFER, INVENT), M25-M45. In this task the simulation system developed in T7.2 as part of the learning resources will become a digital twin, because it will be set up to be populated with real or closest-to-real data relating to the domain of interest. The simulator will eventually be able to incorporate open data or output data from real tests, in order to make the simulations even closer to reality. The process will be broken in two phases: 1. Design, with identification of technologies best suited to enable the real-time flow of data from the demonstrators developed in WP5; 2. Operation: use of the data and models within the twin to perform simulations to help with operational performance of the demonstrators.

T7.4: Modular replication of learning resources (leader: POLIMI; participants: GRIFO, TAU, UPAT, ICAM), M40-M48. The learning resources will be primarily structured and tested in a digital learning path through the MOOC (Massive Open Online Course) platform managed by POLIMI. The MOOC will implement an active approach using a mix of synchronous and asynchronous activities (and tools), such as video lectures, forum discussions, quiz for formative and summative assessments, project work single or in group (asynchronous); theoretical and practical webinars (synchronous). It will be based on videos, but also, images, infographics, and different multimedia contents useful to support the comprehension and the acquisition of knowledge and skills. At the end of this active experience, content experts will be able to: design a course based on a coherent application of the cognitive alignment principles (deep coherence among learning outcomes, assessment strategies, learning activities); integrate in the design of a course the various dimensions of active pedagogical frameworks to improve the achievement of intended learning outcomes. The MOOC will be available on the POLIMI Open Knowledge platform (http://www.pok.polimi.it), currently populated with around 80 courses and 127.000 unique users. All the videos produced for the MOOC will be uploaded on POLIMI Open Knowledge channel (https://www.youtube.com/channel/UCOKtBGblWKlkE-KIf-S0Fvg) too, thus available to the huge audience of the video-sharing platform. Finally, the possibility to include such modules in existing or to-be-developed curricula for undergraduate studies will be assessed, with the involvement of some of the European university networks POLIMI and the other academic partners are currently part of (e.g., ENHANCE, IDEA League, Alliance4Tech).

Work package WP8 – Communication, dissemination and exploitation

Work Package Number	WP8	**Lead Beneficiary**	15. G_MEDIA
Work Package Name	Communication, dissemination and exploitation		
Start Month	1	**End Month**	48

Objectives
Facilitate knowledge exchange between project partners and participants from the project outset; to inform, communicate and engage with the relevant stakeholders and general public concerning all actions and initiatives developed in RECREATE project. This WP will aim: to successfully develop and implement the plan of dissemination of the project results to the public and all relevant stakeholders; to define a strategy for exploitation of results, tools and solutions developed in RECREATE; to deliver and manage IPR knowledge within the Consortium.

Description
T8.1: Public awareness, community engagement, dissemination and communication (leader: G_MEDIA; participants: all), M1-M48. Public Awareness, Community Engagement and Dissemination Strategy will be designed, identifying the stakeholders to target, the modalities to reach and engage them and the other features to ensure a targeted awareness raising and dissemination of the project results and active involvement of the stakeholders. A time-defined Dissemination Action Plan will be designed including the project web site design, implementation and curation. Two internal knowledge exchange meetings will be organised. Two external dissemination events will be organised and four targeted engagement workshops to raise awareness about the project results among specific target groups. Targeted publicity will be assured by the substantial mentions of project results and their potential impact in at least 6 articles in focused publications.

T8.2: Exploitation strategy: identification and validation of key exploitable results (leader: G_MEDIA; participants: all), M1-M18.
The exploitation manager will organise at M12 a dedicated session to present to all partners the tools and the methodologies to address exploitation issues. In parallel, an exploitation workshop will be organised to scout Key Exploitable Results (KERs) generated by the project. For each KER a Lean Canvas will be drafted and discussed among partners during dedicated working sessions. Short working sessions will be organised to progress on the definition of the Lean Canvas for the identified KERs. After the workshop, all partners will be asked to verify such assumptions through the interaction with relevant stakeholders. Once feedbacks are collected from such early validation experience, the Lean Canvas of each KER will be updated by the exploitation manager and results will be discussed at the relevant exploitation workshop. In total, a minimum of 4 exploitation workshops will be organised.

T8.3: Definition and finalisation of the exploitation plan and business plan (leader: G_MEDIA; participants: all), M16-M36.
Outcomes of the previous tasks will be resumed in the RECREATE Exploitation Plan (M36) which will include all exploitation roadmaps for each identified KER. In particular, specific Business Plans will be prepared for the most promising KERs. They will also include specific recommendations on IPR issues. In the final year of the project a workshop on "How to pitch" will be organised to build capacities of partners in presenting project results, key elements of the developed business plans and paving the way to impact, in strong connection with dissemination and communication activities. The exploitation plan and the business plans will be drafted by the Exploitation Manager during the project by considering also outcomes and feedbacks from demonstrators in WP5.

T8.4: Knowledge management and IPR protection (leader: POLIMI; participants: all), M1-M48.
This task will work on (i) creating an appropriate methodology and strategy for knowledge management and protection of intellectual property rights, (ii) informing all the partners of the available IPR protection mechanisms, (iii) organize IPR workshops among the partners for harmonizing IPR definition, (iv) define a plan for boosting the proper use of the IPR on the project foreground knowledge.

T8.5: Legislative framework analysis (Leader: COBAT COMPOSITI; participants: G_MEDIA, AVK, APRA Europe), M34-M48.
In this task, legislation issues will be analysed considering logistics, safety, and environmental implications towards EoL composite reuse and recycling. Specifically, the application of the Extended Producer Responsibility (EPR) principle and the definition of the End of Waste Regulation for composites will be investigated at European level. A position paper with proposals on legislation will be elaborated, with the clear indication of potential benefits associated to proposed modifications. The outcomes of the task will be used as dissemination tools to raise awareness among institutional bodies, such as national and regional authorities, as well as technological clusters. This will intensify the collaboration and synergies between industry, academia and government on this subject, defining measures and market mechanisms to stimulate the development of innovative businesses based on recycling processes, secondary markets and producer responsibility.

Work package WP9 – Project management

Work Package Number	WP9	**Lead Beneficiary**	1. POLIMI
Work Package Name	Project management		
Start Month	1	**End Month**	48

Objectives
Establish the proper communications channels and set-up the necessary control infrastructure in order to provide an adequate project management throughout the project; scientific, technical, financial, balance project management.

Description
T9.1. Financial, legal and administrative coordination activities (leader: POLIMI; participants: all), M1-M48. This task will take care of creating and updating the Project Handbook containing necessary information to run the project on a day-to-day basis and administer the project eRoom integrated within the project website (T8.1). Moreover, it will plan and arrange common RECREATE project meetings (every 4 months in the beginning of the project, every 6 months later in the project) and implement quality procedures for handling of project's quality issues, including the selection of

peer reviewers for reporting documents evaluation following the “Good research practice”. Prof. G. Griffini, as Project Coordinator (PC) and chairman for the General Assembly, will: communicate with Consortium & EC, providing legal & financial information about the partners during the GA preparation phase, resolving administrative or contractual issues during the project; prepare and update the CA, including attention to IPR management, in collaboration with the CDE manager; prepare and post-process EC reviews from the Consortium side, including support in the implementation of any recommendations from the EC and reviewers, with the support of the WP leaders; coordinate payment records, financial accounts, and collect financial statements; manage day-to-day administration of the project; collect deliverable/ milestone reports for submission; supervise preparation of periodic progress reports.

T9.2. Scientific/technical coordination activities and data management (leader: POLIMI; participants: all), M1-M48.
In this task, the PC will: oversee all scientific/technical aspects; manage project progress and achievements vs. scientific and technical objectives (KPIs, KERs); ensure project schedule is met (milestones, deliverables, reports); ensure smooth flow of information between partners. For the management activities, POLIMI will work closely with the General Assembly and WP leaders who will report directly to the PC and be responsible for: workflow scheduling and work plan change control procedure; communication between partners within the WP; coordination of technical activities within the WP; minimization of possible duplication of efforts among partners; valorization of complementary skills and competences within the Consortium. This task will also prepare and continuously update a DMP (Section 1.2.7), with the objective to increase the visibility of public data generated by the project, in compliance with GDPR regulation.

T9.3: Risk assessment and mitigation strategies for the proposed technologies (leader: G_MEDIA; participants: all), M1-M48.
Within RECREATE, risk management will be tackled as a dynamic process, where potential threats are identified, evaluated and controlled via the definition and implementation of appropriate contingency plans (acceptance, transfer, avoidance, mitigation). WP leaders will directly handle technical risk management for each WP under the supervision of the CDE Manager. In particular, in this task, the activities will start from the risk identification and management Table 3.1 and will be focused on (1) risk identification and assessment and (2) mitigation actions at the level of each participating innovative technology concept. The main output will be the creation of a detailed risk identification and management plan updated every 12 months.

T9.4: Monitoring and implementation of project inclusion and transdisciplinarity (leader: POLIMI; participants: all), M1-48.
In this task, the PC supported by all WP leaders will coordinate the efforts in ensuring sex/gender equality, inclusion and professional growth of early-career researchers throughout the project, by means of dedicated focus groups and periodic meetings. Particular attention will be devoted to fostering transdisciplinary activities within the project through the involvement of those partners specializing in the fields of social sciences and humanities (METID@POLIMI, GRIFO), as detailed in Section 1.2.4. An “Inclusion Committee” will be set up by month 6 (1 representative/partner) to oversee such issues in the project.

Work package WP10 – Ethics requirements

Work Package Number	WP10	**Lead Beneficiary**	1. POLIMI
Work Package Name	Ethics requirements		
Start Month	1	**End Month**	48

Objectives
The objective is to ensure compliance with the 'ethics requirements' set out in this work package.

Description
This work package sets out the 'ethics requirements' that the project must comply with.

Associated with document Ref. Ares(2022)3823781 - 20/05/2022

STAFF EFFORT

Staff effort per participant

Grant Preparation (Work packages - Effort screen) — Enter the info.

Participant	WP1	WP2	WP3	WP4	WP5	WP6	WP7	WP8	WP9	WP10	Total Person-Months
1 - POLIMI											
2 - TAU											
3 - Fraunhofer											
4 - ICAM											
5 - UPAT											
6 - GAIKER											
7 - CNRS											
8 - INVENT											
9 - IRIS											
10 - COBAT COMPOSITI											
11 - RESCOLL											
12 - REST UG											
13 - BENASEDO											
14 - Carbon Cleanup											
15 - G_MEDIA											
16 - GRIFO											
17 - EDAG											
18 - HEAD Sport GmbH											
19 - GEVEN SPA											

Staff effort per participant

Grant Preparation (Work packages - Effort screen) — Enter the info.

Participant	WP1	WP2	WP3	WP4	WP5	WP6	WP7	WP8	WP9	WP10	Total Person-Months
20 - APRA Europe											
21 - AVK											
Total Person-Months											

Associated with document Ref. Ares(2022)3823781 - 20/05/2022

LIST OF DELIVERABLES

Deliverables

Grant Preparation (Deliverables screen) — Enter the info.

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D1.1	Mapping and assessment of complex EoL composite waste stream	WP1	10 - COBAT COMPOSITI	R — Document, report	PU - Public	6
D1.2	Technical specifications and target KPIs for final demonstrators	WP1	17 - EDAG	R — Document, report	SEN - Sensitive	9
D1.3	EoL composite part selection	WP1	10 - COBAT COMPOSITI	R — Document, report	SEN - Sensitive	36
D2.1	Smart recognition and monitoring process optimization	WP2	9 - IRIS	R — Document, report	SEN - Sensitive	27
D2.2	Process for automatic evaluation of NDT results	WP2	3 - Fraunhofer	R — Document, report	SEN - Sensitive	36
D2.3	Decision Support System for part reuse	WP2	1 - POLIMI	DEM — Demonstrator, pilot, prototype	SEN - Sensitive	30
D2.4	Digital twin of laser shock dismantling process	WP2	7 - CNRS	DEM — Demonstrator, pilot, prototype	SEN - Sensitive	36
D2.5	Optimization of reshaping process of GFRC EoL parts	WP2	4 - ICAM	R — Document, report	SEN - Sensitive	36
D3.1	Process optimization of electrofragmentation technology for EoL FRC parts	WP3	1 - POLIMI	R — Document, report	SEN - Sensitive	36

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D3.2	Process optimization of catalyst-assisted green solvolysis for EoL FRC parts	WP3	1 - POLIMI	R — Document, report	SEN - Sensitive	36
D3.3	Preparation of rGF and rCF and supply for demonstrators	WP3	1 - POLIMI	DEM — Demonstrator, pilot, prototype	PU - Public	36
D3.4	Reclaimed fibre properties and fibre/matrix characteristics	WP3	2 - TAU	R — Document, report	PU - Public	48
D3.5	Process validation of new recycling technologies	WP3	14 - Carbon Cleanup	R — Document, report	SEN - Sensitive	48
D4.1	Methodology for modular design of next-gen FRC structures	WP4	8 - INVENT	R — Document, report	SEN - Sensitive	30
D4.2	Guidelines for design of reversible joining in next-gen FRC parts	WP4	17 - EDAG	R — Document, report	PU - Public	33
D4.3	Design and development of reversible adhesives and joints	WP4	11 - RESCOLL	R — Document, report	SEN - Sensitive	36
D4.4	Reversible resins for FRC repair and reuse	WP4	1 - POLIMI	R — Document, report	PU - Public	36
D5.1	Demo-case 1 - GFRC reuse	WP5	6 - GAIKER	DEM — Demonstrator, pilot, prototype	SEN - Sensitive	45
D5.2	Demo-case 2 - CFRC reuse	WP5	17 - EDAG	DEM — Demonstrator, pilot, prototype	SEN - Sensitive	45

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D5.3	Demo-case 3 - CFRC reuse	WP5	12 - REST UG	DEM — Demonstrator, pilot, prototype	SEN - Sensitive	45
D5.4	Demo-case 4 - GFRC/CFRC reuse	WP5	19 - GEVEN SPA	DEM — Demonstrator, pilot, prototype	SEN - Sensitive	45
D5.5	Demo-case 5 - rGF/rCFs reprocessing	WP5	19 - GEVEN SPA	DEM — Demonstrator, pilot, prototype	SEN - Sensitive	45
D5.6	Demo-case 6 - smart recognition/monitoring technologies for sorting	WP5	9 - IRIS	DEM — Demonstrator, pilot, prototype	SEN - Sensitive	45
D5.7	Demo-case 7 - rCF reprocessing	WP5	18 - HEAD Sport GmbH	DEM — Demonstrator, pilot, prototype	SEN - Sensitive	45
D5.8	Demo-case 8 - rGF reprocessing	WP5	18 - HEAD Sport GmbH	DEM — Demonstrator, pilot, prototype	SEN - Sensitive	45
D5.9	Demo-case 9 - industrial coatings	WP5	13 - BENASEDO	DEM — Demonstrator, pilot, prototype	SEN - Sensitive	45
D5.10	Intermediate status and assessment of demonstrators	WP5	1 - POLIMI	R — Document, report	SEN - Sensitive	36
D5.11	Final status and assessment of demonstrators	WP5	1 - POLIMI	R — Document, report	PU - Public	48
D6.1	Innovation management programme	WP6	15 - G_MEDIA	R — Document, report	PU - Public	36
D6.2	Circularity assessment tool: methodology and results	WP6	5 - UPAT	OTHER	SEN - Sensitive	48

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D6.3	Preliminary evaluation of EoL reuse/ recycling technologies	WP6	5 - UPAT	R — Document, report	SEN - Sensitive	48
D6.4	LCA/LCC data output	WP6	5 - UPAT	R — Document, report	SEN - Sensitive	42
D6.5	Methodology and results of ESESA	WP6	5 - UPAT	R — Document, report	SEN - Sensitive	48
D7.1	Co-design of relevant teaching methodologies	WP7	1 - POLIMI	R — Document, report	PU - Public	30
D7.2	Specifications and storyboards for learning resources development	WP7	1 - POLIMI	R — Document, report	PU - Public	42
D7.3	Learning resources and simulation software	WP7	16 - GRIFO	OTHER	PU - Public	45
D7.4	Simulation system set up as digital twin	WP7	16 - GRIFO	OTHER	PU - Public	45
D7.5	Validation pathway and evaluation methods	WP7	1 - POLIMI	R — Document, report	PU - Public	45
D7.6	Analysis of results of validation sessions	WP7	1 - POLIMI	R — Document, report	PU - Public	48
D8.1	Public awareness, engagement and dissemination strategy plan	WP8	15 - G_MEDIA	R — Document, report	SEN - Sensitive	3
D8.2	Dissemination action plan	WP8	15 - G_MEDIA	R — Document, report	SEN - Sensitive	4
D8.3	Project website launched and running	WP8	15 - G_MEDIA	DEC —Websites, patent filings, videos, etc	PU - Public	6
D8.4	Key exploitable results	WP8	15 - G_MEDIA	R — Document, report	SEN - Sensitive	18

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D8.5	Exploitation plan	WP8	15 - G_MEDIA	R — Document, report	SEN - Sensitive	36
D8.6	Catalogue of dissemination and exploitation activities	WP8	15 - G_MEDIA	R — Document, report	PU - Public	48
D8.7	IPR information and distribution among partners	WP8	1 - POLIMI	R — Document, report	SEN - Sensitive	48
D8.8	Analysis of legislation framework	WP8	10 - COBAT COMPOSITI	R — Document, report	PU - Public	48
D9.1	Quality assurance plan	WP9	1 - POLIMI	R — Document, report	SEN - Sensitive	4
D9.2	Economic and technical progress report	WP9	1 - POLIMI	R — Document, report	SEN - Sensitive	12
D9.3	Data management plan	WP9	1 - POLIMI	R — Document, report	SEN - Sensitive	6
D9.4	Risk assessment and management plan	WP9	15 - G_MEDIA	R — Document, report	SEN - Sensitive	6
D9.5	Project inclusion and transdisciplinarity	WP9	1 - POLIMI	R — Document, report	PU - Public	12
D10.1	OEI - Requirement No. 1	WP10	1 - POLIMI	ETHICS	SEN - Sensitive	1

Deliverable – Mapping and assessment of complex EoL composite waste stream

Deliverable Number	D1.1	**Lead Beneficiary**	10. COBAT COMPOSITI
Deliverable Name	Mapping and assessment of complex EoL composite waste stream		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	6	**Work Package No**	WP1

Description
This deliverable will report the work of task T1.1 concerning the analysis, mapping and assessment of the EoL composite waste material streams in Europe in terms of composition, volumes, availability and locations, covering different relevant industrial sectors.

Deliverable – Technical specifications and target KPIs for final demonstrators

Deliverable Number	D1.2	**Lead Beneficiary**	17. EDAG
Deliverable Name	Technical specifications and target KPIs for final demonstrators		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	9	**Work Package No**	WP1

Description
This deliverable will report the work carried out in Task T1.2 and will include the industrial requirements as well as the target Key Performance Indicators for the project demo-cases.

Deliverable – EoL composite part selection

Deliverable Number	D1.3	**Lead Beneficiary**	10. COBAT COMPOSITI
Deliverable Name	EoL composite part selection		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP1

Description
This deliverable will report the work carried out in T1.3. It will schematically describe the EoL composite parts to be used in the technical developments of WP2,3 and in the demonstrators (WP5), on the basis of the proposed form compiled by each technological partner.

Deliverable – Smart recognition and monitoring process optimization

Deliverable Number	D2.1	**Lead Beneficiary**	9. IRIS
Deliverable Name	Smart recognition and monitoring process optimization		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	27	**Work Package No**	WP2

Description

This deliverable will report the work of task T2.1 concerning the smart recognition and monitoring techniques. As main results the most convenient technology configuration and associated AI models will be developed and other evaluation tools identified.

Deliverable – Process for automatic evaluation of NDT results

Deliverable Number	D2.2	**Lead Beneficiary**	3. Fraunhofer
Deliverable Name	Process for automatic evaluation of NDT results		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP2

Description
This deliverable will report the work of task T2.2 concerning different non-destructive approaches for inspecting EoL composite parts. The different technologies are used for efficient defect detection, characterization and repair. Decision on the fate of the EoL FRC components will be made possible regarding the potential for part reuse or recycling. The result is presented in the form of a report.

Deliverable – Decision Support System for part reuse

Deliverable Number	D2.3	**Lead Beneficiary**	1. POLIMI
Deliverable Name	Decision Support System for part reuse		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	SEN - Sensitive
Due Date (month)	30	**Work Package No**	WP2

Description
This deliverable will report the work of Task T2.3. It will contain the description, preliminary validation and demonstration of a DSS for part reuse, able to acquire and elaborate information and requirements for the different components and assemblies.

Deliverable – Digital twin of laser shock dismantling process

Deliverable Number	D2.4	**Lead Beneficiary**	7. CNRS
Deliverable Name	Digital twin of laser shock dismantling process		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP2

Description
This deliverable will report the work of task T2.4 concerning the development and optimization of a dismantling process using laser wave shock technology with double and symmetrical impact configurations. The demonstrator will consist in a digital twin model of such process.

Deliverable – Optimization of reshaping process of GFRC EoL parts

Deliverable Number	D2.5	**Lead Beneficiary**	4. ICAM
Deliverable Name	Optimization of reshaping process of GFRC EoL parts		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP2

Description
This deliverable will report the work of task T2.5 concerning reshaping of used polyester glass fiber composites. The goals are to define the conditions and limits of a process able to modify parts in shape to make them available to other uses, to develop a process able to handle large parts minimizing the number of processing steps.

Deliverable – Process optimization of electrofragmentation technology for EoL FRC parts

Deliverable Number	D3.1	**Lead Beneficiary**	1. POLIMI
Deliverable Name	Process optimization of electrofragmentation technology for EoL FRC parts		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP3

Description
This deliverable will report the work of Task 3.1. It will contain the description, preliminary validation and experimental demonstration of a new efficient size-reduction process based on high-voltage fragmentation, reporting the extensive experimental campaign and the optimization procedure.

Deliverable – Process optimization of catalyst-assisted green solvolysis for EoL FRC parts

Deliverable Number	D3.2	**Lead Beneficiary**	1. POLIMI
Deliverable Name	Process optimization of catalyst-assisted green solvolysis for EoL FRC parts		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP3

Description
This deliverable will report the work of task T3.2 concerning the development and optimization of a mild solvolysis process for depolymerization of FRCs through an extensive dedicated experimental campaign.

Deliverable – Preparation of rGF and rCF and supply for demonstrators

Deliverable Number	D3.3	**Lead Beneficiary**	1. POLIMI
Deliverable Name	Preparation of rGF and rCF and supply for demonstrators		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP3

Description
This deliverable will report the work of task T3.1 an T3.2 concerning the reclamation of rGF and rCF from electrofragmentation and catalyst-assisted low-temperature green solvolysis. The output will be the supply of rGF and RCF for demonstrators.

Deliverable – Reclaimed fibre properties and fibre/matrix characteristics

Deliverable Number	D3.4	**Lead Beneficiary**	2. TAU
Deliverable Name	Reclaimed fibre properties and fibre/matrix characteristics		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP3

Description
This deliverable will report the work of task T3.3. The focus is on the characterization of short individual recycled fibers before and after re-sizing.

Deliverable – Process validation of new recycling technologies

Deliverable Number	D3.5	**Lead Beneficiary**	14. Carbon Cleanup
Deliverable Name	Process validation of new recycling technologies		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	48	**Work Package No**	WP3

Description
This deliverable will report the work of task T3.4 concerning the benchmarking activities to qualify and validate the recycling technologies developed in RECREATE. The report will include the results of such activities in terms of key validation parameters focusing on sustainability, process and economic factors.

Deliverable – Methodology for modular design of next-gen FRC structures

Deliverable Number	D4.1	**Lead Beneficiary**	8. INVENT
Deliverable Name	Methodology for modular design of next-gen FRC structures		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	30	**Work Package No**	WP4

Description
This deliverable will report the work of task T4.1 concerning modular design of next-gen composite structures. Approaches for a modular design will be derived from the use cases (wind rotor blades, aircraft interior, automotive parts etc.). Generalizations will be made from the use cases in order to finally establish generally applicable design rules.

Deliverable – Guidelines for design of reversible joining in next-gen FRC parts

Deliverable Number	D4.2	**Lead Beneficiary**	17. EDAG

Deliverable Name	Guidelines for design of reversible joining in next-gen FRC parts		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	33	**Work Package No**	WP4

Description
This deliverable will report the work of task T4.2 concerning the design of reversible joining solutions as a key factor for reuse of components and recycling. The report will deal with different adhesives and their possible performance if considering not only the use phase, but also the dismantling of a product.

Deliverable – Design and development of reversible adhesives and joints

Deliverable Number	D4.3	**Lead Beneficiary**	11. RESCOLL
Deliverable Name	Design and development of reversible adhesives and joints		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP4

Description
This deliverable will report the work of task T4.3 concerning the design and development of reversible adhesives and joints. The report will include a detailed description of the properties of the developed materials and the characterization of their performance.

Deliverable – Reversible resins for FRC repair and reuse

Deliverable Number	D4.4	**Lead Beneficiary**	1. POLIMI
Deliverable Name	Reversible resins for FRC repair and reuse		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP4

Description
This deliverable will report the work of task T4.4 concerning the design and development of reversible resins to be used as matrix materials in FRCs for repair and reuse. The report will include a detailed characterization of the properties of the selected materials and of their performance.

Deliverable – Demo-case 1 - GFRC reuse

Deliverable Number	D5.1	**Lead Beneficiary**	6. GAIKER
Deliverable Name	Demo-case 1 - GFRC reuse		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	SEN - Sensitive
Due Date (month)	45	**Work Package No**	WP5

Description

This deliverable will report the work of task T5.1 and will contain the results of demo-case 1. Reshaped flat panels will be used to develop technical boards for the transport sector validating the reshaping technology developed in T2.5.

Deliverable – Demo-case 2 - CFRC reuse

Deliverable Number	D5.2	**Lead Beneficiary**	17. EDAG
Deliverable Name	Demo-case 2 - CFRC reuse		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	SEN - Sensitive
Due Date (month)	45	**Work Package No**	WP5

Description
This deliverable will report the work of task T5.2 and will contain the results of demo-case 2. The idea of the democase is to design a vehicle structure with a maximised content of reusable components utilizing the technologies developed in the previous workpackages. The demonstrator will be used to validate the technologies and to showcase them in exhibitions.

Deliverable – Demo-case 3 - CFRC reuse

Deliverable Number	D5.3	**Lead Beneficiary**	12. REST UG
Deliverable Name	Demo-case 3 - CFRC reuse		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	SEN - Sensitive
Due Date (month)	45	**Work Package No**	WP5

Description
This deliverable will report the work of task T5.3 and will contain the results of demo-case 3. The target will be the physical demonstration of a new designed rotor-blade for wind turbines with high content of reused CFRC.

Deliverable – Demo-case 4 - GFRC/CFRC reuse

Deliverable Number	D5.4	**Lead Beneficiary**	19. GEVEN SPA
Deliverable Name	Demo-case 4 - GFRC/CFRC reuse		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	SEN - Sensitive
Due Date (month)	45	**Work Package No**	WP5

Description
This deliverable will report the work of task T5.4 and will contain the results of demo-case 4. The focus will be on the demonstration of the use of EoL GFRC/CFRC parts in panels for aircraft seat components and cabin interiors.

Deliverable – Demo-case 5 - rGF/rCFs reprocessing

Deliverable Number	D5.5	**Lead Beneficiary**	19. GEVEN SPA

Deliverable Name	Demo-case 5 - rGF/rCFs reprocessing		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	SEN - Sensitive
Due Date (month)	45	**Work Package No**	WP5

Description
This deliverable will report the work of task T5.5 and will contain the results of demo-case 5. The focus will be on the demonstration of the use of rGFs/rCFs in cabin lining panels for the passenger Cabin and Service Area in aircrafts.

Deliverable – Demo-case 6 - smart recognition/monitoring technologies for sorting

Deliverable Number	D5.6	**Lead Beneficiary**	9. IRIS
Deliverable Name	Demo-case 6 - smart recognition/monitoring technologies for sorting		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	SEN - Sensitive
Due Date (month)	45	**Work Package No**	WP5

Description
This deliverable will report the work of task T5.6 and will contain the results of demo-case 6. The focus will be on the demonstration of a recognition monitoring prototype unit for FRC EoL parts.

Deliverable – Demo-case 7 - rCF reprocessing

Deliverable Number	D5.7	**Lead Beneficiary**	18. HEAD Sport GmbH
Deliverable Name	Demo-case 7 - rCF reprocessing		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	SEN - Sensitive
Due Date (month)	45	**Work Package No**	WP5

Description
This deliverable will report the work of task T5.7 and will contain the results of demo-case 7. The focus will be on the demonstration of the use of rCF to produce parts of rackets with flat surface.

Deliverable – Demo-case 8 - rGF reprocessing

Deliverable Number	D5.8	**Lead Beneficiary**	18. HEAD Sport GmbH
Deliverable Name	Demo-case 8 - rGF reprocessing		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	SEN - Sensitive
Due Date (month)	45	**Work Package No**	WP5

Description

This deliverable will report the work of task T5.8 and will contain the results of demo-case 8. The focus will be on the demonstration of the use of rGFs in the structural architecture of amateur skis.

Deliverable – Demo-case 9 - industrial coatings

Deliverable Number	D5.9	**Lead Beneficiary**	13. BENASEDO
Deliverable Name	Demo-case 9 - industrial coatings		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	SEN - Sensitive
Due Date (month)	45	**Work Package No**	WP5

Description
This deliverable will report the work of task T5.9 and will contain the results of demo-case 9. The focus will be on the demonstration of the use of the organic fraction recovered by solvolytic experiments (T3.2) as base component for industrial coating formulations.

Deliverable – Intermediate status and assessment of demonstrators

Deliverable Number	D5.10	**Lead Beneficiary**	1. POLIMI
Deliverable Name	Intermediate status and assessment of demonstrators		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP5

Description
This deliverable reports the intermediate output of task T5.10 and presents the ongoing status of the nine RECREATE demonstrators.

Deliverable – Final status and assessment of demonstrators

Deliverable Number	D5.11	**Lead Beneficiary**	1. POLIMI
Deliverable Name	Final status and assessment of demonstrators		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP5

Description
This deliverable will report the work of task T5.10 related to the analysis of the final status of the nine RECREATE demonstrators.

Deliverable – Innovation management programme

Deliverable Number	D6.1	**Lead Beneficiary**	15. G_MEDIA
Deliverable Name	Innovation management programme		
Type	R — Document, report	**Dissemination Level**	PU - Public

Due Date (month)	36	**Work Package No**	WP6

Description
This deliverable will report the work of task T6.1 regarding the technology concepts that will be verified in a controlled environment and designed in innovation workshops. The report will include information on the component and breadboard validation in relevant environment that will be carried out for each tested technology. A circularity assessment tool will be developed and applied.

Deliverable – Circularity assessment tool: methodology and results

Deliverable Number	D6.2	**Lead Beneficiary**	5. UPAT
Deliverable Name	Circularity assessment tool: methodology and results		
Type	OTHER	**Dissemination Level**	SEN - Sensitive
Due Date (month)	48	**Work Package No**	WP6

Description
This deliverable will report the work of T6.2 concerning the development of dedicated digital tools such as the Internet of Things (IoT), big data and data analytics that will be used by the demo-case leaders/end-users to accelerate circular technologies.

Deliverable – Preliminary evaluation of EoL reuse/recycling technologies

Deliverable Number	D6.3	**Lead Beneficiary**	5. UPAT
Deliverable Name	Preliminary evaluation of EoL reuse/recycling technologies		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	48	**Work Package No**	WP6

Description
This deliverable will report the work of task T6.2 concerning the preliminary evaluation of the available EoL reuse/recycling technologies for composite materials. The outcome from the evaluation will be the state-of-the-art upon which RECREATE will build new reuse/recycling technologies.

Deliverable – LCA/LCC data output

Deliverable Number	D6.4	**Lead Beneficiary**	5. UPAT
Deliverable Name	LCA/LCC data output		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	42	**Work Package No**	WP6

Description
This deliverable will report the work of task T6.3 concerning the output data that will be derived from the LCA and LCC analyses of the reused and recycled composite parts.

Deliverable – Methodology and results of ESESA

Deliverable Number	D6.5	**Lead Beneficiary**	5. UPAT
Deliverable Name	Methodology and results of ESESA		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	48	**Work Package No**	WP6

Description
This deliverable will report the work that will be performed in T6.3 concerning the methodology and results of the ESESA on the reused and recycled composite parts.

Deliverable – Co-design of relevant teaching methodologies

Deliverable Number	D7.1	**Lead Beneficiary**	1. POLIMI
Deliverable Name	Co-design of relevant teaching methodologies		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	30	**Work Package No**	WP7

Description
This deliverable will report the work of task T7.1 concerning the co-design of innovative learning/teaching methods in the field of composite circularity. The report will be focused on the definition of relevant Guidelines for Teaching Methodologies and Strategies for the RECREATE project.

Deliverable – Specifications and storyboards for learning resources development

Deliverable Number	D7.2	**Lead Beneficiary**	1. POLIMI
Deliverable Name	Specifications and storyboards for learning resources development		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	42	**Work Package No**	WP7

Description
This deliverable will report the work of task T7.1 concerning the co-design of innovative learning/teaching methods in the field of composite circularity. The report will include a set of storyboards to be used as foundation for the innovative learning system developed in RECREATE.

Deliverable – Learning resources and simulation software

Deliverable Number	D7.3	**Lead Beneficiary**	16. GRIFO
Deliverable Name	Learning resources and simulation software		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	45	**Work Package No**	WP7

Description

This deliverable will report on T7.2 and will consist of the learning resources aimed at the different categories of learners identified (current and future professionals, undergraduates) on topics related to composite circularity. The learning resources will be delivered through a gamification platform.

Deliverable – Simulation system set up as digital twin

Deliverable Number	D7.4	**Lead Beneficiary**	16. GRIFO
Deliverable Name	Simulation system set up as digital twin		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	45	**Work Package No**	WP7

Description
This deliverable will report the work of task T7.3 and will consist of a simulation system developed as part of the learning resources, to be used as a digital twin set up to be populated with real or closest-to-real data relating to the domain of interest.

Deliverable – Validation pathway and evaluation methods

Deliverable Number	D7.5	**Lead Beneficiary**	1. POLIMI
Deliverable Name	Validation pathway and evaluation methods		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	45	**Work Package No**	WP7

Description
This deliverable will report the work of task T7.4 concerning the modular replication and validation of the learning resources developed in RECREATE. The report will include a set of evaluation methods for the learning resources and their validation through the MOOC platform managed by POLIMI.

Deliverable – Analysis of results of validation sessions

Deliverable Number	D7.6	**Lead Beneficiary**	1. POLIMI
Deliverable Name	Analysis of results of validation sessions		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP7

Description
This deliverable will report the work of task T7.4 concerning the modular replication and validation of the learning resources developed in RECREATE. The report will include the output of the validation sessions of the learning resources operated through the MOOC platform managed by POLIMI.

Deliverable – Public awareness, engagement and dissemination strategy plan

Deliverable Number	D8.1	**Lead Beneficiary**	15. G_MEDIA

Deliverable Name	Public awareness, engagement and dissemination strategy plan		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	3	**Work Package No**	WP8

Description
This deliverable will report the work for task T8.1 concerning public awareness, engagement and dissemination strategy plan. It will consist in a preliminary report detailing the planned activities (M3) which will be updated at M12, M24 and M36. The final report delivered at M48 will include the results and implementation of the public awareness, engagement and dissemination strategy in the RECREATE project and an overview of the overall success of the strategy.

Deliverable – Dissemination action plan

Deliverable Number	D8.2	**Lead Beneficiary**	15. G_MEDIA
Deliverable Name	Dissemination action plan		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	4	**Work Package No**	WP8

Description
This deliverable will report the work of task T8.1 concerning the Dissemination Action Plan and the results of its implementation. It will consist of a report released at M4, and updated at M13, M25 and M37. The final report will be delivered at M48 and will include the results of the dissemination action strategy implemented throughout the RECREATE project.

Deliverable – Project website launched and running

Deliverable Number	D8.3	**Lead Beneficiary**	15. G_MEDIA
Deliverable Name	Project website launched and running		
Type	DEC —Websites, patent filings, videos, etc	**Dissemination Level**	PU - Public
Due Date (month)	6	**Work Package No**	WP8

Description
This deliverable will report the work of task T8.1 concerning the project website setup, lunch and monitoring of the website. The outcomes will include website design as well as its implementation and curation.

Deliverable – Key exploitable results

Deliverable Number	D8.4	**Lead Beneficiary**	15. G_MEDIA
Deliverable Name	Key exploitable results		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	18	**Work Package No**	WP8

Description

This deliverable will report the work of task T8.2. It will consist of a report detailing the results of the workshops aimed at the definition of the key exploitable results of the RECREATE project.

Deliverable – Exploitation plan

Deliverable Number	D8.5	**Lead Beneficiary**	15. G_MEDIA
Deliverable Name	Exploitation plan		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP8

Description
This deliverable will report the progress of task T8.3. It will consist of a report including a structured exploitation plan as well as a Lean Canvas for each of the Key Exploitable Results (KERs) identified in T8.2.

Deliverable – Catalogue of dissemination and exploitation activities

Deliverable Number	D8.6	**Lead Beneficiary**	15. G_MEDIA
Deliverable Name	Catalogue of dissemination and exploitation activities		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP8

Description
This deliverable will report the work of task T8.3. The outcome of this deliverable will be a Catalogue of all dissemination and exploitation activities taking place during the RECREATE project including all internal and external workshops as well as dissemination events.

Deliverable – IPR information and distribution among partners

Deliverable Number	D8.7	**Lead Beneficiary**	1. POLIMI
Deliverable Name	IPR information and distribution among partners		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	48	**Work Package No**	WP8

Description
This deliverable will report the work of task T8.4. It will consist of a report including the IPR management plan for the project.

Deliverable – Analysis of legislation framework

Deliverable Number	D8.8	**Lead Beneficiary**	10. COBAT COMPOSITI
Deliverable Name	Analysis of legislation framework		
Type	R — Document, report	**Dissemination Level**	PU - Public

Due Date (month)	48	**Work Package No**	WP8

Description
This deliverable will report the work of task T8.5. It will consist of a position paper on legislation to be used as awareness tool among the public authorities.

Deliverable – Quality assurance plan

Deliverable Number	D9.1	**Lead Beneficiary**	1. POLIMI
Deliverable Name	Quality assurance plan		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	4	**Work Package No**	WP9

Description
This deliverable will report the work of task T9.1. It will consist of a report including the plan of the quality assurance mechanisms established in the project for enabling to achieve the highest quality standards in the produced reports.

Deliverable – Economic and technical progress report

Deliverable Number	D9.2	**Lead Beneficiary**	1. POLIMI
Deliverable Name	Economic and technical progress report		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	12	**Work Package No**	WP9

Description
This deliverable will report the work of task T9.1. It will consist of a report including information on the economic and technical progress of the project. The first version will be released at M12 and updated yearly at M24, M36 and M48.

Deliverable – Data management plan

Deliverable Number	D9.3	**Lead Beneficiary**	1. POLIMI
Deliverable Name	Data management plan		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	6	**Work Package No**	WP9

Description
This deliverable will detail the work of task T9.2 and will consist of a report including the Data Management Plan (DMP). The first version will be released at M6 and further updates will be delivered at M24, M36 and M48.

Deliverable – Risk assessment and management plan

Deliverable Number	D9.4	**Lead Beneficiary**	15. G_MEDIA

Deliverable Name	Risk assessment and management plan		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	6	**Work Package No**	WP9

Description
This deliverable will report the work of task T9.3 concerning risk assessment and mitigation strategies for the proposed technologies. The main output will be the creation of a detailed risk identification and management plan, including the mitigation actions taken during the RECREATE project. The first release will be at M6 and further updates will be at M24, M36 and M48.

Deliverable – Project inclusion and transdisciplinarity

Deliverable Number	D9.5	**Lead Beneficiary**	1. POLIMI
Deliverable Name	Project inclusion and transdisciplinarity		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	12	**Work Package No**	WP9

Description
This deliverable will report the work of task T9.4 and will consist of a report detailing the activities of the "Inclusion Committee". The first release of the report will be at M12 and further updates will be at M24, M36 and M48.

Deliverable – OEI - Requirement No. 1

Deliverable Number	D10.1	**Lead Beneficiary**	1. POLIMI
Deliverable Name	OEI - Requirement No. 1		
Type	ETHICS	**Dissemination Level**	SEN - Sensitive
Due Date (month)	1	**Work Package No**	WP10

Description
As the Consortium did not address any Ethics issues that could arise from project activities, the appointment of an Ethics Advisor is mandatory in order to assure that all Ethics aspects of the proposal are properly taken into account. The independent ethics advisor should have experience in the understanding of AI ethics, health and safety aspects, and personal data processing. The Ethics Advisor should be appointed before the envisioned activities start in WP2 (at M1). The Ethics Advisor should help the Consortium with implementation of the above Recommendations, and an ethics report from the Ethics Advisor is to be submitted at every reporting period.

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LIST OF MILESTONES

Milestones

Grant Preparation (Milestones screen) — Enter the info.

Milestone No	Milestone Name	Work Package No	Lead Beneficiary	Means of Verification	Due Date (month)
1	Definition of input sectors for EoL composite procurement	WP1	10-COBAT COMPOSITI	Document delivered and approved by partners.	12
2	Assessment of laser-shock dismantling process	WP2	7-CNRS	Performance of dismantled parts validated in WP3.	12
3	Assessment of long-term stability of reshaped GFRC parts	WP2	4-ICAM	Flat geometry retention validated after at least 6 months.	18
4	Selection of at least two matrix-fiber pairs for effective electrofragmentation	WP3	1-POLIMI	Quality of liberated fibers validated experimentally in WP3.	24
5	Selection of at least two matrix-catalyst pairs for effective green solvolysis	WP3	1-POLIMI	Quality of liberated fibers validated experimentally in WP3.	24
6	Definition of modular design rules for next-gen composite structures completed	WP4	8-INVENT	Report on design guidelines approved by partners.	28
7	Selection of at least two matrix materials for FRC formulation, repair and reuse	WP4	1-POLIMI	Matrix reversibility validated experimentally in WP3.	28
8	Final assessment of KPIs for demo-cases	WP5	1-POLIMI	Report on KPIs reviewed and approved by partners.	48
9	Release of a digital tool for circularity assessment	WP6	5-UPAT	Software resource available to partners.	36
10	Release of e-learning resources	WP7	16-GRIFO	Software resource available to partners.	45
11	Release of the exploitation roadmap	WP8	15-G_MEDIA	Report on exploitation roadmap approved by partners.	32
12	Set up of the Inclusion Committee	WP9	1-POLIMI	Minutes of 1st online meeting of Inclusion Committee approved by partners.	6

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LIST OF CRITICAL RISKS

Critical risks & risk management strategy *Grant Preparation (Critical Risks screen) — Enter the info.*			
Risk number	**Description**	**Work Package No(s)**	**Proposed Mitigation Measures**
1	Insufficient or difficult supply of EoL composite parts.	WP1	Outsourcing of virgin composite parts and conditioning through accelerated aging cycles to simulate EoL behaviour.
2	Insufficient recognition ability in LIBS.	WP2	Combination of LIBS with other available photonic technologies like Near Infrared (NIR).
3	Limited retention of permanent shape in GFRC reshaped parts.	WP2	Technology validated coupling the thermal treatment with solvochemical treatment to soften the PEs matrix and mimize elastic recovery.
4	Underperforming laser-based dismantling and cut on thicker parts.	WP2	Alternative mechanical cutting technologies will be integrated to the laser treatment.
5	Poor energy efficiency in electrofragmentation.	WP3	Process optimization through DOE chemometric approach and realization of a cyberphysical model.
6	Poor fiber liberation efficiency in solvochemical process for some types of crosslinked resins.	WP3	Thermochemical methods will be adopted to recover fibers and organic liquid fraction (facility already available at POLIMI).
7	Poor adhesion performance in reversible adhesives.	WP4	The nature of the polymeric binder (epoxy, polyurethane, polyamide) will be optimized and tailored surface treatments will be applied.
8	Insufficient availability of recycled material for the realization of all demonstrators.	WP5	Optimization of recycled material loading in combination with virgin material in the final demonstrator for their effective validation.
9	Remanufactured products underperforming vs. original KPIs.	WP5	Target application and associated technical requirements for remanufactured parts will be revised and KPIs updated.
10	Not satisfying economic viability of processes, costs higher than expected.	WP6	Financial/risk analysis performed to validate impact in current and future scenarios, with realignment of circular business models.
11	Limited participation of external stakeholders to the co-design of learning resources.	WP7	Co-design process undertaken with participation of existing networks of professional contacts (e.g., POLIMI study programme advisory boards).

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PROJECT REVIEWS

Project Reviews

Grant Preparation (Reviews screen) — Enter the info.

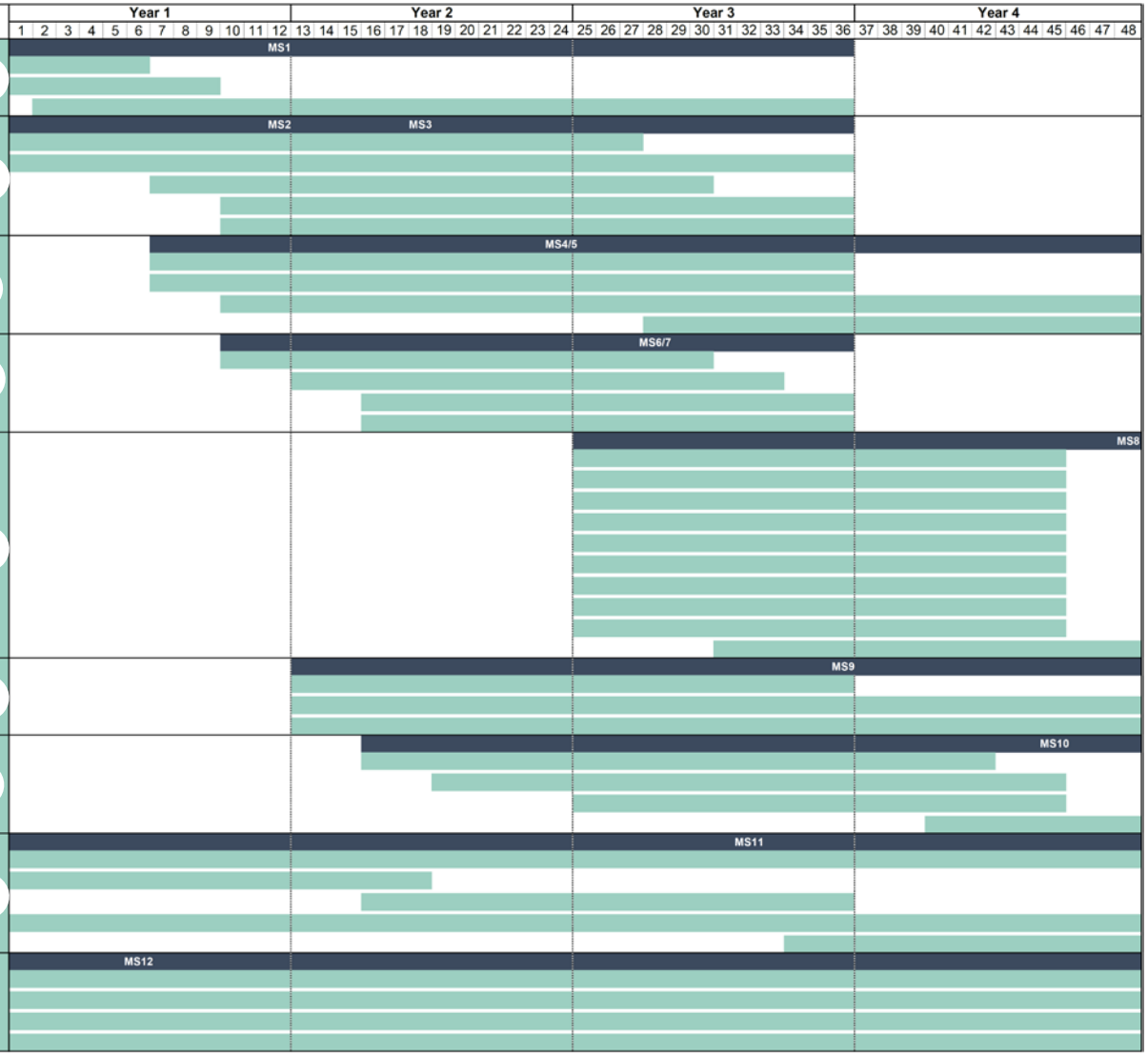
Review No	**Timing** (month)	**Location**	**Comments**
RV1	18		
RV2	36		
RV3	48		

History of changes

Date	n°	Page	Content of the change with respect to the proposal
Excellence			
09.02.2022	1	2	Update of Table 1 with incorporation of additional objectives, expected results and KPI in connection with novel recycling technologies developed in the project.
Impact			
04.04.2022	2	26	Additional details included in Table 12 in relation to the dissemination and communication activities of project results.
Implementation			
09.02.2022	3	Annex 1 - Part A	Additional details on the role and activities of the project coordinator in relation to the management of complementary expertise within the Consortium (Task 9.2).
09.02.2022	4	Annex 1 - Part A	Typos corrected.
09.02.2022	5	Annex 1 - Part A	Further details added on activities related to EoL composite part supply (T3.1) and legislative framework analysis (T8.5).
09.02.2022	6	Annex 1 - Part A	Timing of deliverables in WP4, WP6, WP7 and WP8 adjusted to be in line with associated task duration.
09.02.2022	7	Annex 1 - Part A	Additional deliverables included in WP5 and WP8 to ensure more effective monitoring of project advances.
01.04.2022	8	32	Gantt chart edited to include correct short names of beneficiaries.
04.04.2022	9	Annex 1 - Part A	Additional typos corrected in WP9.
04.04.2022	10	Annex 1 - Part A	In WP9, “Steering committee” substituted with “General Assembly”, in line with governance structure defined in Grant Agreement.
04.04.2022	11	33-34	Additional, more detailed cost justification for partners UPAT; GAIKER; CNRS; IRIS; COBAT COMPOSITI; RESCOLL; REST UG; G_MEDIA.
Additional changes			
01.04.2022	12	2-35 and Annex 1 – Part A	Short names of beneficiaries made consistent throughout the text.
20.04.2022	13	34	Revision of cost justification for partner G_MEDIA.

Table 4 – Gantt chart of the RECREATE project.

RECREATE	Leader	Start	End
WP1 Input EoL streams and output requirements	**COBAT COMPOSITI**	**1**	**36**
T1.1 Mapping of suitable EoL composite waste	COBAT COMPOSITI	1	6
T1.2 Definition of technical specifications for final demonstrators	EDAG	1	9
T1.3 Selection and supply of EoL parts	COBAT COMPOSITI	2	36
WP2 Re-use and remanufacturing of EoL composite structures	**FRAUNHOFER**	**1**	**36**
T2.1 Smart recognition and monitoring techniques for sorting	IRIS	1	27
T2.2 Non-destructive inspection, defect characterization and repair	FRAUNHOFER	1	36
T2.3 Decision Support System (DSS) for part re-use	POLIMI	7	30
T2.4 Active dismantling of complex and multilayer structures	CNRS	10	36
T2.5 Temperature-assisted reshaping	ICAM	10	36
WP3 Recycling and reprocessing of materials from non-reusable parts	**POLIMI**	**7**	**48**
T3.1 Electrofragmentation	POLIMI	7	36
T3.2 Catalyst-assisted low-temperature green solvolysis	POLIMI	7	36
T3.3 Materials characterization, preparation and reprocessing	TAU	10	48
T3.4 Benchmarking and process validation of new recycling technologies	CARBON CLEANUP	28	48
WP4 Next-generation recyclable-by-design composites	**INVENT**	**10**	**36**
T4.1 Modular design of next-gen composite structures	INVENT	10	30
T4.2 Design and requirements for reversible joining	EDAG	13	33
T4.3 Reversible adhesives and joints	RESCOLL	16	36
T4.4 Reversible green resins for multiple reuse, repair and recycling	POLIMI	16	36
WP5 Design and development of demonstrators	**EDAG**	**25**	**48**
T5.1 Reuse of EoL GFRC parts in technical boards for transportation	GAIKER	25	45
T5.2 Reuse of EoL CFRC parts in modular structures for mobility	EDAG	25	45
T5.3 Reuse of EoL CFRC parts in segmentable rotor blades for wind energy	REST UG	25	45
T5.4 Reuse of EoL GFRC/CFRC parts for aircraft cabin interiors	GEVEN SPA	25	45
T5.5 Reprocessing of recycled CFs and next-gen resins for aircraft cabin interiors	GEVEN SPA	25	45
T5.6 Validation of smart recognition and monitoring technologies for sorting	IRIS	25	45
T5.7 Reprocessing of recycled CFs and next-gen resins for sport equipment	HEAD SPORT GMBH	25	45
T5.8 Reprocessing of recycled GFs for sport equipment	HEAD SPORT GMBH	25	45
T5.9 Validation of recycled organic fraction for industrial coatings	BENASEDO	25	45
T5.10 Monitoring and assessment of demonstrators against KPIs	POLIMI	31	48
WP6 Circular economy and sustainability assessment	**UPAT**	**13**	**48**
T6.1 Innovation management and circular business plans	G_MEDIA	13	36
T6.2 Digital tools for the demonstration and acceleration of circularity	UPAT	13	48
T6.3 Integrated environmental and socio-economic sustainability assessment	UPAT	13	48
WP7 Co-design and validation of learning resources	**GRIFO**	**16**	**48**
T7.1 Definition of innovative learning/teaching methods	POLIMI	16	42
T7.2 Technological implementation of learning resources within a game environmer	GRIFO	19	45
T7.3 Digital twins of key technologies	GRIFO	25	45
T7.4 Modular replication of learning resources	POLIMI	40	48
WP8 Communication, dissemination and exploitation	**G_MEDIA**	**1**	**48**
T8.1 Public awareness, community engagement, dissemination and communicatior	G_MEDIA	1	48
T8.2 Exploitation strategy: identification and validation of key exploitable results	G_MEDIA	1	18
T8.3 Definition and finalisation of the exploitation plan and business plan	G_MEDIA	16	36
T8.4 Knowledge management and IPR protection	POLIMI	1	48
T8.5 Legislative framework analysis	COBAT COMPOSITI	34	48
WP9 Project management	**POLIMI**	**1**	**48**
T9.1 Financial, legal and administrative coordination activities	POLIMI	1	48
T9.2 Scientific/technical coordination activities and data management	POLIMI	1	48
T9.3 Risk assessment and mitigation strategies for the proposed technologies	G_MEDIA	1	48
T9.4 Monitoring and implementation of project inclusion and transdisciplinarity	POLIMI	1	48



In Table 3.1h, “Purchase costs” item is described for participants where such costs exceed 15% of personnel costs.

Table 3.1h. ‘Purchase costs’ items (travel and subsistence, equipment and other goods, works and services).

5/UPAT	Cost (€)	Justification
Travel and subsistence	20’000	8 technical progress meetings (2’000€/person); 2 international conferences (2’000€/person).
Equipment	0	-
Other goods, works and services	10’000	3-year license (7’000) for the GaBi Circularity Toolkit to be used to perform LCA and LCC and to determine the Material Circularity Indicator. The Toolkit will be used 100% for the purposes of RECREATE. Payment of open access publications fees (2 x 1’500€).
Total	30’000	
6/GAIKER	**Cost (€)**	**Justification**
Travel and subsistence	15’000	Participation in 8 consortium meetings (2 people/meeting: 13’000€), dissemination events (1 people/ event: 2’000€, conference).
Equipment	8’000	Depreciation costs for equipment for composite manufacturing (compression molding), 100% used for the project.
Other goods, works and services	32’000	Specific moulds (800x800mm2), tubes, and membranes for reshaping activities (WP5) (25’000€), resins, solvents, reinforcements (WP3/WP5) (7’000€).
Total	55’000	
7/CNRS	**Cost (€)**	**Justification**
Travel and subsistence	15’000	Participation (2 people) in 8 project meetings/2 review meetings (12’000€), conference events (3’000€).
Equipment	0	-
Other goods, works and services	60’000	Optics for laser focalization (lenses, mirrors, electronics), handling adaptation for samples, maintenance/calibration of laser units and robot, lab consumables (glassware, chemicals for analyses), sensors for laser shock adhesion test (LASAT) online implementation, digital twin and software license.
Total	75’000	
9/IRIS	**Cost (€)**	**Justification**
Travel and subsistence	28’000	Participation (2 people) in 8 project meetings and 2 review meetings (20k€), 2 technical meetings (4k€), 2 conference events (4k€).
Equipment	0	-
Other goods, works and services	180’600	(i) Materials for fabrication of the prototype for the monitoring system (key components/pieces needed for prototype construction): cameras (100k€); optomechanical parts for sampling (20k€); software licenses (QT Creator (IDE), Matlab+PLS toolbox (modelling), Google Drive (data storage), TIA portal (automation)), IT Servers for hosting and deployment digital monitoring system platform + hardware, laptop, pc, cables, connections/adapters, hard disk (25k€); sample analysis (15k€) // (ii) IPR and patents (8k€) // (iii) Publications (5k€) // (iv) shipment and insurance costs for the prototype (4K€) // (v) Audit (3.6k€).
Total	208’600	
10/COBAT COMPOSITI	**Cost (€)**	**Justification**
Travel and subsistence	22’500	Participation (2 people) in kick-off meeting, 4 yearly meetings and 4 intermediate consortium meetings
Equipment	0	-
Other goods, works and services	40’000	Supply of different types of EoL wind blade formulations (GF/CF), according to the specific experimental requirements (only in case of unavailability, insufficient or difficult supply of a requested blade type within the Cobat Compositi network, the parts will be purchased externally; see Risk n.1, Table 3.1e). Logistics costs (including collection, in-situ preparation/cutting, storage, delivery) for the composite parts’ shipment to the different partners. EoL management of the residual parts/granulates from experiments.
Total	62’500	
11/RESCOLL	**Cost (€)**	**Justification**
Travel and subsistence	12’000	Participation in 8 projects meeting (2 people/meeting), participation in 2-4 dissemination and communication events (conferences, workshops).
Equipment	41’056	Depreciation costs for equipment for adhesive deposition and laser shock equipment: bonding deposition system (24 months, 25% use rate); mixing system (24 months, 25% use rate); automated bonding deposition system (24 months, 25% use rate); rheology lab equipment (24 months, 25% use rate); Dispermat (24 months, 25% use rate); laser system (36 months, 50% use rate).
Other goods, works and services	38’500	Chemicals for formulation of adhesives (resins, hardeners, solvents). Mixing vessels & lids, cartridges, dispensing disks, seals for the mixers, substrates for bonding tests, small lab equipment (beakers, analyses chemicals, heating plates), parts for triggering equipment (heating patches). Laser consumables equipment (optics, lens, mirrors, electronics).
Total	91’556	

12/REST UG	Cost (€)	Justification
Travel and subsistence	10’000	Participation in 8 technical progress meetings (1 person).
Equipment	0	-
Other goods, works and services	40’000	License fee software (Ansys finite element analysis for structural calculations - 100% used for the project), tools for demonstrator production (WP5), transportation of wind blade segments.
Total	50’000	

15/G_MEDIA	Cost (€)	Justification
Travel and subsistence	14’400	2 team members/12 meetings, seminars and workshops planned for CDE activities.
Equipment	0	-
Other goods, works and services	54’200	50’000€ events organization costs (12 events of various types); 4’200€ print and production costs. Type and number of events: 2 technical meetings (7’000€ per event); 6 seminars and workshops (3’600€ per event); 4 working group meetings (3’600€ per event). Events costs include: venue and equipment costs; costs of refreshments for participants.
Total	68’600	

3.2 Capacity of participants and consortium as a whole

The RECREATE consortium consists of 21 partners from 9 different EU countries (Fig.17). Overall, the consortium is highly multidisciplinary and exhibits very complementary expertise and competences. The lead partner is **Politecnico di Milano** (POLIMI), which is the 1st public technical university in Italy, the 6th in Europe in the area “Engineering” (https://www.topuniversities.com/universities/politecnico-di-milano). POLIMI will participate with 3 different groups, namely the Department of Chemistry, Materials and Chemical Engineering “Giulio Natta” (https://www.cmic.polimi.it, with project coordinator Prof. G. Griffini, green solvolysis, smart resins), the Department of Mechanical Engineering (https://www.mecc.polimi.it, electrofragmentation, AI-based decision support systems), and METID (https://www.metid.polimi.it/), which will be engaged in the co-design and realization of innovative learning resources in collaboration with GRIFO, for the realization of MOOCs, serious gaming approaches to composite circularity education and digital twins (https://www.grifomultimedia.it/en/home-en/). POLIMI has a well-established and documented experience as participant and coordinator of EU-funded research projects, and researchers from the two Departments are commonly engaged in several projects concerning circular economy, composite recycling, and smart polymers. Moreover, POLIMI will have a direct access to some of the European university networks POLIMI and the other academic partners are currently part of (e.g., ENHANCE, IDEA League, Alliance4Tech), for an efficient replication of the innovative learning tools developed in the project.

The other research partners involved in RECREATE as technology providers are:

Fig. 17 Geographical distribution of RECREATE

- **Tampere University** (TAU), with recognized expertise in polymers and composites characterization (https://www.tuni.fi/en/research/plastics-and-elastomer-technology), and with unique specialized facilities for single-fiber mechanical testing (https://fibrobotics.com/);
- **Fraunhofer IWU and WKI** (FRAUNHOFER), leading the activities of design for disassembly aimed at the reuse of various EoL composite parts. FRAUNHOFER labs are well equipped with different NDT technologies such as active/passive thermography, X-ray Computer Tomography, ultrasound monitoring and radar-based technologies, that will be crucial for the assessment of defects and for the selection of reusable EoL parts. Information about FRAUNH capabilities and instrumental facilities are available at: https://ati.ec.europa.eu/technology-centre/fraunhofer-institute-machine-tools-and-forming-technology-fraunhofer-iwu; https://www.iwu.fraunhofer.de/de/forschung/leistungsangebot/kompetenzen-von-a-bis-z/nachhaltigkeit-und-circular-economy.html;
- **Icam Ouest** (ICAM), in charge of the activities dedicate to thermal reshaping of GFRCs, as a follow up of recent internal positive achievements (https://en.icam.fr/innovative-structures-and-materials/);
- **University of Patras** (UPAT), expert in the development of dedicated digital tools, IoT, big data analysis, socio-economic sustainability assessment and LCA/LCC with particular reference to the case of composite circular economy;
- **Centre national de la recherche scientifique** (CNRS), expert in laser-based technologies (https://pimm.artsetmetiers.fr/en/equipes/laser-laser-processes), will lead all the activities related to composite part dismantling and machining for repair. A link demonstrating available laser facilities is available at https://youtu.be/J9jUu_7bFTs

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The role of key-technology providers will be completed by **Iris Technology Development** (IRIS, see https://www.iris-eng.com/), that will develop an innovative recognition and sorting technology based on laser induced breakdown spectroscopy (LIBS). The company is already owner of AI-based NIR technology for fast recognition and sorting: VISUM InLine, an Industrial NIR analyzer for real-time and continuous chemical composition monitoring adapted to several types of geometries and unit operations and unit process machinery.

Rescoll (RESCOLL) and **Invent** (INVENT, see https://www.invent-gmbh.de/en/unternehmen/forschung/, https://www.invent-gmbh.de/en/unternehmen/fertigung/, https://www.invent-gmbh.de/en/unternehmen/produktentwicklung/) will be the responsible for the design and development of modular composite structures based on detachable joints and dedicated reversible adhesives.

Some of the research partners (POLIMI, FRAUNHOFER, TAU) are partners of the ***Vanguard Initiative*** (https://www.s3vanguardinitiative.eu/) and already cooperating in other common projects about circular economy. This will help to guarantee a smooth start of the project without delays, an efficient monitoring of the advancement of the activities, and an effective platform for targeted dissemination and stakeholder engagement actions.

Two **recycling SMEs**, namely **Cobat Compositi** (COBAT COMPOSITI) and **Carbon Clean-Up** (Carbon Cleanup), will play an important twofold role in the project:

- they will select and provide suitable sizes, geometries and amounts of EoL composite parts to the other project partners, after a careful mapping of available waste at EU level;
- they will validate the new technologies and products developed in the frame of the project, performing a techno-economical evaluation of the competitivity of the proposed solution in comparison to what is currently available in the market (e.g., recycled CF from pyrolysis).

Realization of **demonstrators**, integrating all the key technologies developed in the project, will be assigned to both **SMEs** like **REST UG** and **BENASEDO**, and **large companies like EDAG engineering**, **GEVEN SPA** (https://www.geven.com/official-home-page/test-laboratories/) and **HEAD Sport GmbH**. All these companies possess dedicated testing facilities where demonstration of the case studies developed in the project will be carried out in relevant industrial environment (TRL6). One of the demonstration activities focused on remanufacturing and reuse of large polyester/vinylester GFRC materials will be led by **Gaiker** (GAIKER), a deep-tech RTO dedicated to transfer relevant technologies to the market. Within this framework, GAIKER will implement up to TRL6 the T-assisted reshaping approach developed at lower TRL(3) by ICAM during the project, thus ultimately improving composites circularity in relevant industrial sectors (mobility, transportation).

The industrial partners will be supported in the formalization of appropriate sustainable circular business cases by **Giacomelli-Media** (G_MEDIA), which possess relevant expertise in delivering international training programs for companies in energy and circular economy sectors (see www.circularbusinessacademy.com). G_MEDIA will be also the responsible partner for the dissemination activities, where they will be supported by sectoral associations focused on composites and remanufacturing (**AVK** and **APRA Europe**, respectively). G_MEDIA possess a vast and diverse network of industrial contacts including central-east Europe, and could engage stakeholders potentially very interested to the results of RECREATE project.

Finally, **dedicated competences in humanities and social sciences** are guaranteed by the presence of **METID** in the Consortium, with the support of **GRIFO Multimedia** for gamification and realization of relevant digital tools, as well as by the activities undertaken by UPAT for the development of an integrated environmental, social and economic sustainability assessment. Dedicated activities concerning Open Sciences and Gender inclusion will be continuously monitored by POLIMI, which will be leading a task in WP9 (T9.4) fully dedicated to these aspects.

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ANNEX 2a

ADDITIONAL INFORMATION ON UNIT COSTS AND CONTRIBUTIONS

SME owners/natural person beneficiaries without salary (Decision C(2020) 7115[1])

Type: unit costs

Units: days spent working on the action (rounded up or down to the nearest half-day)

Amount per unit (daily rate): calculated according to the following formula:

{EUR 5 080 / 18 days = **282,22**}
multiplied by
{country-specific correction coefficient of the country where the beneficiary is established}

The country-specific correction coefficients used are those set out in the Horizon Europe Work Programme (section Marie Skłodowska-Curie actions) in force at the time of the call (see Portal Reference Documents).

HE and Euratom Research Infrastructure actions[2]

Type: unit costs

Units[3]: see (for each access provider and installation) the unit cost table in Annex 2b

Amount per unit*: see (for each access provider and installation) the unit cost table in Annex 2b

* Amount calculated as follows:

For trans-national access:

$$\frac{\text{average annual total trans-national access costs to the installation (over past two years[4])}}{\text{average annual total quantity of trans-national access to the installation (over past two years[5])}}$$

For virtual access:

$$\frac{\text{total virtual access costs to the installation (over the last year[6])}}{\text{total quantity of virtual access to the installation (over the last year[7])}}$$

Euratom staff mobility costs[8]

Monthly living allowance

Type: unit costs

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1 Commission Decision of 20 October 2020 authorising the use of unit costs for the personnel costs of the owners of small and medium-sized enterprises and beneficiaries that are natural persons not receiving a salary for the work carried out by themselves under an action or work programme (C(2020)7715).

2 Decision of 19 April 2021 authorising the use of unit costs for the costs of providing trans-national and virtual access in Research Infrastructure actions under the Horizon Europe Programme (2021-2027) and the Research and Training Programme of the European Atomic Energy Community (2021-2025).

3 Unit of access (e.g. beam hours, weeks of access, sample analysis) fixed by the access provider in proposal.

4 In exceptional and duly justified cases, the granting authority may agree to a different reference period.

5 In exceptional and duly justified cases, the granting authority may agree to a different reference period.

6 In exceptional and duly justified cases, the granting authority may agree to a different reference period.

7 In exceptional and duly justified cases, the granting authority may agree to a different reference period.

8 Decision of 15 March 2021 authorising the use of unit costs for mobility in co-fund actions under the Research and Training Programme of the European Atomic Energy Community (2021-2025).

Units: months spent by the seconded staff member(s) on research and training in fission and fusion activities (person-month)

Amount per unit*: see (for each beneficiary/affiliated entity and secondment) the unit cost table in Annex 2b

* Amount calculated as follows from 1 January 2021:
{**EUR 4 300** multiplied by
country-specific correction coefficient** of the country where the staff member is seconded}[9]

**Country-specific correction coefficients as from 1 January 2021[10]

EU-Member States[11]

Country / Place	Coefficient (%)
Bulgaria	59,1
Czech Rep.	85,2
Denmark	131,3
Germany Bonn Karlsruhe Munich	101,9 95,8 98 113,9
Estonia	82,3
Ireland	129
Greece	81,4
Spain	94,2
France	120,5
Croatia	75,8
Italy Varese	95 90,7
Cyprus	78,2
Latvia	77,5
Lithuania	76,6
Hungary	71,9
Malta	94,7
Netherlands	113,9
Austria	107,9
Poland	70,9
Portugal	91,1
Romania	66,6
Slovenia	86,1

[9] Unit costs for living allowances are calculated by using a method of calculation similar to that applied for the secondment to the European Commission of seconded national experts (SNEs).

[10] For the financial statements, the amount must be adjusted according to the actual place of secondment.
The revised coefficients were adopted in the Decision authorising the use of unit costs for the Fusion Programme co-fund action under the Research and training Programme of the European Atomic Energy Community 2021-2025. They are based on the 2020 Annual update of the remuneration and pensions of the officials and other servants of the European Union and the correction coefficients applied thereto (OJ C 428, 11.12.2020) to ensure purchasing power parity. The revised coefficient are applied as from 1 January 2021 through an amendment to the grant agreement.

[11] No correction coefficient shall be applicable in Belgium and Luxembourg.

Slovakia	80,6
Finland	118,4
Sweden	124,3

Third countries

Country/place	Coefficient (%)
China	82,2
India	72,3
Japan	111,8
Russia	92,7
South Korea	92,3
Switzerland	129.2
Ukraine	82.3
United Kingdom	97.6
United States	101,4 (New-York) 90,5 (Washington)

Mobility allowance

Type: Unit costs

Units: months spent by the seconded staff member(s) on research and training in fission and fusion activities (person-month)

Amount per unit: **EUR 600** per person-month; see (for each beneficiary/affiliated entity and secondment) the unit cost table in Annex 2b

Family allowance

Type: unit costs

Units: months spent by the seconded staff member(s) on research and training in fission and fusion activities (person-month)

Amount per unit: **EUR 660** per person-month; see (for each beneficiary/affiliated entity and secondment) the unit cost table in Annex 2b

Education allowance

Type: Unit costs

Units: months spent by the seconded staff member(s) on research and training in fission and fusion activities (person-month)

Amount per unit*: see (for each beneficiary/affiliated entity and secondment) the unit cost table in Annex 2b

*Amount calculated as follows from 1 January 2021:
{EUR 283.82 x number of dependent children[12]**}**

[12] For the estimated budget (Annex 2): an average should be used. (⚠ For the financial statements, the number of children (and months) must be adjusted according to the actual family status at the moment the secondment starts.)

Associated with document Ref. Ares(2022)3823781 - 20/05/2022

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

TAMPEREEN KORKEAKOULUSAATIO SR (TAU), PIC 902999288, established in KALEVANTIE 4, TAMPERE 33100, Finland,

hereby agrees

to become beneficiary

in Agreement No 101058756 — RECREATE ('the Agreement')

between POLITECNICO DI MILANO (POLIMI) **and** the **European Health and Digital Executive Agency (HADEA)** ('EU executive agency' or 'granting authority'), under the powers delegated by the European Commission ('European Commission'),

and mandates

the coordinator to submit and sign in its name and on its behalf any **amendments** to the Agreement, in accordance with Article 39.

By signing this accession form, the beneficiary accepts the grant and agrees to implement it in accordance with the Agreement, with all the obligations and terms and conditions it sets out.

SIGNATURE

For the beneficiary

Associated with document Ref. Ares(2022)3823781 - 20/05/2022

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

FRAUNHOFER GESELLSCHAFT ZUR FORDERUNG DER ANGEWANDTEN FORSCHUNG EV (Fraunhofer), PIC 999984059, established in HANSASTRASSE 27C, MUNCHEN 80686, Germany,

hereby agrees

to become beneficiary

in Agreement No 101058756 — RECREATE ('the Agreement')

between POLITECNICO DI MILANO (POLIMI) **and** the **European Health and Digital Executive Agency (HADEA)** ('EU executive agency' or 'granting authority'), under the powers delegated by the European Commission ('European Commission'),

and mandates

the coordinator to submit and sign in its name and on its behalf any **amendments** to the Agreement, in accordance with Article 39.

By signing this accession form, the beneficiary accepts the grant and agrees to implement it in accordance with the Agreement, with all the obligations and terms and conditions it sets out.

SIGNATURE

For the beneficiary

Associated with document Ref. Ares(2022)3823781 - 20/05/2022

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

ICAM OUEST (ICAM), PIC 998210996, established in 35 CHE DU CHAMP DE MANOEUVRES, CARQUEFOU 44470, France,

hereby agrees

to become beneficiary

in Agreement No 101058756 — RECREATE ('the Agreement')

between POLITECNICO DI MILANO (POLIMI) **and** the **European Health and Digital Executive Agency (HADEA)** ('EU executive agency' or 'granting authority'), under the powers delegated by the European Commission ('European Commission'),

and mandates

the coordinator to submit and sign in its name and on its behalf any **amendments** to the Agreement, in accordance with Article 39.

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SIGNATURE

For the beneficiary

Associated with document Ref. Ares(2022)3823781 - 20/05/2022

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

PANEPISTIMIO PATRON (UPAT), PIC 999894528, established in UNIVERSITY CAMPUS RIO PATRAS, RIO PATRAS 265 04, Greece,

hereby agrees

to become beneficiary

in Agreement No 101058756 — RECREATE ('the Agreement')

between POLITECNICO DI MILANO (POLIMI) **and** the **European Health and Digital Executive Agency (HADEA)** ('EU executive agency' or 'granting authority'), under the powers delegated by the European Commission ('European Commission'),

and mandates

the coordinator to submit and sign in its name and on its behalf any **amendments** to the Agreement, in accordance with Article 39.

By signing this accession form, the beneficiary accepts the grant and agrees to implement it in accordance with the Agreement, with all the obligations and terms and conditions it sets out.

SIGNATURE

For the beneficiary

Associated with document Ref. Ares(2022)3823781 - 20/05/2022

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

FUNDACION GAIKER (GAIKER), PIC 999615071, established in PARQUE TECNOLOGICO DE VIZCAYA, EDIFICIO 202, ZAMUDIO 48170, Spain,

hereby agrees

to become beneficiary

in Agreement No 101058756 — RECREATE ('the Agreement')

between POLITECNICO DI MILANO (POLIMI) **and** the **European Health and Digital Executive Agency (HADEA)** ('EU executive agency' or 'granting authority'), under the powers delegated by the European Commission ('European Commission'),

and mandates

the coordinator to submit and sign in its name and on its behalf any **amendments** to the Agreement, in accordance with Article 39.

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SIGNATURE

For the beneficiary

Associated with document Ref. Ares(2022)3823781 - 20/05/2022

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

CENTRE NATIONAL DE LA RECHERCHE SCIENTIFIQUE CNRS (CNRS), PIC 999997930, established in RUE MICHEL ANGE 3, PARIS 75794, France,

hereby agrees

to become beneficiary

in Agreement No 101058756 — RECREATE ('the Agreement')

between POLITECNICO DI MILANO (POLIMI) **and** the **European Health and Digital Executive Agency (HADEA)** ('EU executive agency' or 'granting authority'), under the powers delegated by the European Commission ('European Commission'),

and mandates

the coordinator to submit and sign in its name and on its behalf any **amendments** to the Agreement, in accordance with Article 39.

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SIGNATURE

For the beneficiary

Associated with document Ref. Ares(2022)3823781 - 20/05/2022

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

INVENT INNOVATIVE VERBUNDWERKSTOFFEREALISATION UND VERMARKTUNG NEUERTECHNOLOGIEN GMBH (INVENT), PIC 998801629, established in CHRISTIAN POMMER STRASSE 47, BRAUNSCHWEIG 38112, Germany,

hereby agrees

to become beneficiary

in Agreement No 101058756 — RECREATE ('the Agreement')

between POLITECNICO DI MILANO (POLIMI) **and** the **European Health and Digital Executive Agency (HADEA)** ('EU executive agency' or 'granting authority'), under the powers delegated by the European Commission ('European Commission'),

and mandates

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By signing this accession form, the beneficiary accepts the grant and agrees to implement it in accordance with the Agreement, with all the obligations and terms and conditions it sets out.

SIGNATURE

For the beneficiary

Associated with document Ref. Ares(2022)3823781 - 20/05/2022

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

IRIS TECHNOLOGY SOLUTIONS, SOCIEDAD LIMITADA (IRIS), PIC 998993786, established in CARRETERA D ESPLUGUES 39 41, CORNELLA DE LLOBREGAT 08940, Spain,

hereby agrees

to become beneficiary

in Agreement No 101058756 — RECREATE ('the Agreement')

between POLITECNICO DI MILANO (POLIMI) **and** the **European Health and Digital Executive Agency (HADEA)** ('EU executive agency' or 'granting authority'), under the powers delegated by the European Commission ('European Commission'),

and mandates

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SIGNATURE

For the beneficiary

Associated with document Ref. Ares(2022)3823781 - 20/05/2022

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

COBAT COMPOSITI (COBAT COMPOSITI), PIC 889348963, established in VIA VICENZA 29, ROMA 00185, Italy,

hereby agrees

to become beneficiary

in Agreement No 101058756 — RECREATE ('the Agreement')

between POLITECNICO DI MILANO (POLIMI) **and** the **European Health and Digital Executive Agency (HADEA)** ('EU executive agency' or 'granting authority'), under the powers delegated by the European Commission ('European Commission'),

and mandates

the coordinator to submit and sign in its name and on its behalf any **amendments** to the Agreement, in accordance with Article 39.

By signing this accession form, the beneficiary accepts the grant and agrees to implement it in accordance with the Agreement, with all the obligations and terms and conditions it sets out.

SIGNATURE

For the beneficiary

Associated with document Ref. Ares(2022)3823781 - 20/05/2022

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

RESCOLL (RESCOLL), PIC 998135821, established in ALLEE GEOFFROY SAINT HILAIRE 8 CS 30021, PESSAC 33600, France,

hereby agrees

to become beneficiary

in Agreement No 101058756 — RECREATE ('the Agreement')

between POLITECNICO DI MILANO (POLIMI) **and** the **European Health and Digital Executive Agency (HADEA)** ('EU executive agency' or 'granting authority'), under the powers delegated by the European Commission ('European Commission'),

and mandates

the coordinator to submit and sign in its name and on its behalf any **amendments** to the Agreement, in accordance with Article 39.

By signing this accession form, the beneficiary accepts the grant and agrees to implement it in accordance with the Agreement, with all the obligations and terms and conditions it sets out.

SIGNATURE

For the beneficiary

Associated with document Ref. Ares(2022)3823781 - 20/05/2022

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

REST UG (REST UG), PIC 896759957, established in WESTERENDE 20, SCHWABSTEDT 25876, Germany,

hereby agrees

to become beneficiary

in Agreement No 101058756 — RECREATE ('the Agreement')

between POLITECNICO DI MILANO (POLIMI) **and** the **European Health and Digital Executive Agency (HADEA)** ('EU executive agency' or 'granting authority'), under the powers delegated by the European Commission ('European Commission'),

and mandates

the coordinator to submit and sign in its name and on its behalf any **amendments** to the Agreement, in accordance with Article 39.

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SIGNATURE

For the beneficiary

Associated with document Ref. Ares(2022)3823781 - 20/05/2022

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

N. BENASEDO SPA (BENASEDO), PIC 984710439, established in VIA ANGUISSOLA SOFONISBA 2, Milano 20100, Italy,

hereby agrees

to become beneficiary

in Agreement No 101058756 — RECREATE ('the Agreement')

between POLITECNICO DI MILANO (POLIMI) **and** the **European Health and Digital Executive Agency (HADEA)** ('EU executive agency' or 'granting authority'), under the powers delegated by the European Commission ('European Commission'),

and mandates

the coordinator to submit and sign in its name and on its behalf any **amendments** to the Agreement, in accordance with Article 39.

By signing this accession form, the beneficiary accepts the grant and agrees to implement it in accordance with the Agreement, with all the obligations and terms and conditions it sets out.

SIGNATURE

For the beneficiary

Associated with document Ref. Ares(2022)3823781 - 20/05/2022

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

CARBON CLEANUP GMBH (Carbon Cleanup), PIC 892032953, established in PETER-BEHRENS-PLATZ 9, LINZ 4020, Austria,

hereby agrees

to become beneficiary

in Agreement No 101058756 — RECREATE ('the Agreement')

between POLITECNICO DI MILANO (POLIMI) **and** the **European Health and Digital Executive Agency (HADEA)** ('EU executive agency' or 'granting authority'), under the powers delegated by the European Commission ('European Commission'),

and mandates

the coordinator to submit and sign in its name and on its behalf any **amendments** to the Agreement, in accordance with Article 39.

By signing this accession form, the beneficiary accepts the grant and agrees to implement it in accordance with the Agreement, with all the obligations and terms and conditions it sets out.

SIGNATURE

For the beneficiary

Associated with document Ref. Ares(2022)3823781 - 20/05/2022

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

GIACOMELLI MEDIA MANAGEMENT IN SVETOVANJE DOO (G_MEDIA), PIC 914861709, established in KERSNIKOVA ULICA 12, LJUBLJANA 1000, Slovenia,

hereby agrees

to become beneficiary

in Agreement No 101058756 — RECREATE ('the Agreement')

between POLITECNICO DI MILANO (POLIMI) **and** the **European Health and Digital Executive Agency (HADEA)** ('EU executive agency' or 'granting authority'), under the powers delegated by the European Commission ('European Commission'),

and mandates

the coordinator to submit and sign in its name and on its behalf any **amendments** to the Agreement, in accordance with Article 39.

By signing this accession form, the beneficiary accepts the grant and agrees to implement it in accordance with the Agreement, with all the obligations and terms and conditions it sets out.

SIGNATURE

For the beneficiary

Associated with document Ref. Ares(2022)3823781 - 20/05/2022

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

GRIFO MULTIMEDIA SRL (GRIFO), PIC 942558119, established in VIA GALILEO GALILEI 15, RUVO DI PUGLIA 70037, Italy,

hereby agrees

to become beneficiary

in Agreement No 101058756 — RECREATE ('the Agreement')

between POLITECNICO DI MILANO (POLIMI) **and** the **European Health and Digital Executive Agency (HADEA)** ('EU executive agency' or 'granting authority'), under the powers delegated by the European Commission ('European Commission'),

and mandates

the coordinator to submit and sign in its name and on its behalf any **amendments** to the Agreement, in accordance with Article 39.

By signing this accession form, the beneficiary accepts the grant and agrees to implement it in accordance with the Agreement, with all the obligations and terms and conditions it sets out.

SIGNATURE

For the beneficiary

Associated with document Ref. Ares(2022)3823781 - 20/05/2022

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

EDAG ENGINEERING GMBH (EDAG), PIC 927231634, established in KREUZBERGER RING 40, WIESBADEN 65205, Germany,

hereby agrees

to become beneficiary

in Agreement No 101058756 — RECREATE ('the Agreement')

between POLITECNICO DI MILANO (POLIMI) **and** the **European Health and Digital Executive Agency (HADEA)** ('EU executive agency' or 'granting authority'), under the powers delegated by the European Commission ('European Commission'),

and mandates

the coordinator to submit and sign in its name and on its behalf any **amendments** to the Agreement, in accordance with Article 39.

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SIGNATURE

For the beneficiary

Associated with document Ref. Ares(2022)3823781 - 20/05/2022

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

HEAD SPORT GMBH (HEAD Sport GmbH), PIC 929476602, established in WUHRKOPFWEG 1, KENNELBACH 6921, Austria,

hereby agrees

to become beneficiary

in Agreement No 101058756 — RECREATE ('the Agreement')

between POLITECNICO DI MILANO (POLIMI) **and** the **European Health and Digital Executive Agency (HADEA)** ('EU executive agency' or 'granting authority'), under the powers delegated by the European Commission ('European Commission'),

and mandates

the coordinator to submit and sign in its name and on its behalf any **amendments** to the Agreement, in accordance with Article 39.

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SIGNATURE

For the beneficiary

Associated with document Ref. Ares(2022)3823781 - 20/05/2022

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

GEVEN SPA (GEVEN SPA), PIC 986282033, established in ZONA INDUSTRIALE ASI/ MARIGLIANO, LOC BOSCOFANGONE, NOLA 80035, Italy,

hereby agrees

to become beneficiary

in Agreement No 101058756 — RECREATE ('the Agreement')

between POLITECNICO DI MILANO (POLIMI) **and** the **European Health and Digital Executive Agency (HADEA)** ('EU executive agency' or 'granting authority'), under the powers delegated by the European Commission ('European Commission'),

and mandates

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SIGNATURE

For the beneficiary

Associated with document Ref. Ares(2022)3823781 - 20/05/2022

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

AUTOMOTIVE PARTS REMANUFACTURERS ASSOCIATION EUROPE (APRA Europe), PIC 933341858, established in TERVURENLAAN 34, BRUSSELS 1040, Belgium,

hereby agrees

to become beneficiary

in Agreement No 101058756 — RECREATE ('the Agreement')

between POLITECNICO DI MILANO (POLIMI) **and** the **European Health and Digital Executive Agency (HADEA)** ('EU executive agency' or 'granting authority'), under the powers delegated by the European Commission ('European Commission'),

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SIGNATURE

For the beneficiary

Associated with document Ref. Ares(2022)3823781 - 20/05/2022

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

AVK-INDUSTRIEVEREINIGUNG VERSTARKTEKUNSTSTOFFE EV (AVK), PIC 998597735, established in AM HAUPTBAHNHOF 10, FRANKFURT AM MAIN 60329, Germany,

hereby agrees

to become beneficiary

in Agreement No 101058756 — RECREATE ('the Agreement')

between POLITECNICO DI MILANO (POLIMI) **and** the **European Health and Digital Executive Agency (HADEA)** ('EU executive agency' or 'granting authority'), under the powers delegated by the European Commission ('European Commission'),

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SIGNATURE

For the beneficiary

FINANCIAL STATEMENT FOR [PARTICIPANT NAME] FOR REPORTING PERIOD [NUMBER]

	Eligible[1] costs (per budget category)																	EU contribution[2]				Revenues
	Direct costs															Indirect costs	Total costs	EU contribution to eligible costs			Total requested EU contribution	Income generated by the action
	A. Personnel costs			B. Subcontracting costs	C. Purchase costs			D. Other cost categories								E. Indirect costs[2]		Funding rate %[3]	Maximum EU contribution[4]	Requested EU contribution		
	A.1 Employees (or equivalent) A.2 Natural persons under direct contract A.3 Seconded persons		A.4 SME owners and natural person beneficiaries	B. Subcontracting	C.1 Travel and subsistence	C.2 Equipment	C.3 Other goods, works and services	[D.1 Financial support to third parties]	D.2 Internally invoiced goods and services	[D.3 Transnational access to research infrastructure unit costs]	[D.4 Virtual access to research infrastructure unit costs]	*[OPTION for HE PCP/PPI:* D.5 PCP/PPI procurement costs*]*	*[OPTION for Euratom Programme Cofund Actions:* D.6 Euratom Cofund staff mobility costs*]*	*[OPTION for HE ERC Grants:* D.7 ERC additional funding*]*	*[OPTION for HE ERC Grants:* D.8 ERC additional funding (subcontracting, FSTP and internally invoiced goods and services)*]*	E. Indirect costs						
Forms of funding	Actual costs	Unit costs (usual accounting practices)	Unit costs[5]	Actual costs	Actual costs	Actual costs	Actual costs	[Actual costs]	Unit costs (usual accounting practices)	[Unit costs[5]]	[Unit costs[5]]	[Actual costs]	[Unit costs[5]]	[Actual costs]	[Actual costs]	Flat-rate costs[6]						
	a1	a2	a3	b	c1	c2	c3	[d1a]	d2	[d3]	[d4]	[d5]	[d6]	[d7]	[d8]	e = 0,25 * (a1 + a2 + a3 ~~+ b~~ + c1 +c2 + c3 ~~+ d1a + d2 + d3 + d4 + d5[+ d6]~~ [+d7] ~~[+ d8]~~)	f = a+b+c+d+e	U	g = f*U%	h	m	n
XX – [short name beneficiary/affiliated entity]																						

The beneficiary/affiliated entity hereby confirms that:

The information provided is complete, reliable and true.

The costs and contributions declared are eligible (see Article 6).

The costs and contributions can be substantiated by adequate records and supporting documentation that will be produced upon request or in the context of checks, reviews, audits and investigations (see Articles 19, 20 and 25).

For the last reporting period: that all the revenues have been declared (see Article 22).

ⓘ Please declare all eligible costs and contributions, even if they exceed the amounts indicated in the estimated budget (see Annex 2). Only amounts that were declared in your individual financial statements can be taken into account lateron, in order to replace costs/contributions that are found to be ineligible.

[1] See Article 6 for the eligibility conditions. All amounts must be expressed in EUR (see Article 21 for the conversion rules).

[2] If you have also received an EU operating grant during this reporting period, you cannot claim indirect costs - unless you can demonstrate that the operating grant does not cover any costs of the action. This requires specific accounting tools. Please contact us immediately via the Funding & Tenders Portal for details.

[3] See Data Sheet for the reimbursement rate(s).

[4] This is the *theoretical* amount of EU contribution to costs that the system calculates automatically (by multiplying the reimbursement rates by the costs declared). The amount you request (in the column 'requested EU contribution') may be less.

[5] See Annex 2a 'Additional information on the estimated budget' for the details (units, cost per unit).

[6] See Data Sheet for the flat-rate.

ANNEX 5

SPECIFIC RULES

CONFIDENTIALITY AND SECURITY (— ARTICLE 13)

Sensitive information with security recommendation

Sensitive information with a security recommendation must comply with the additional requirements imposed by the granting authority.

Before starting the action tasks concerned, the beneficiaries must have obtained all approvals or other mandatory documents needed for implementing the task. The documents must be kept on file and be submitted upon request by the coordinator to the granting authority. If they are not in English, they must be submitted together with an English summary.

For requirements restricting disclosure or dissemination, the information must be handled in accordance with the recommendation and may be disclosed or disseminated only after written approval from the granting authority.

EU classified information

If EU classified information is used or generated by the action, it must be treated in accordance with the security classification guide (SCG) and security aspect letter (SAL) set out in Annex 1 and Decision 2015/444[1] and its implementing rules — until it is declassified.

Deliverables which contain EU classified information must be submitted according to special procedures agreed with the granting authority.

Action tasks involving EU classified information may be subcontracted only with prior explicit written approval from the granting authority and only to entities established in an EU Member State or in a non-EU country with a security of information agreement with the EU (or an administrative arrangement with the Commission).

EU classified information may not be disclosed to any third party (including participants involved in the action implementation) without prior explicit written approval from the granting authority.

ETHICS (— ARTICLE 14)

Ethics and research integrity

The beneficiaries must carry out the action in compliance with:

- ethical principles (including the highest standards of research integrity)

[1] Commission Decision 2015/444/EC, Euratom of 13 March 2015 on the security rules for protecting EU classified information (OJ L 72, 17.3.2015, p. 53).

and

- applicable EU, international and national law, including the EU Charter of Fundamental Rights and the European Convention for the Protection of Human Rights and Fundamental Freedoms and its Supplementary Protocols.

No funding can be granted, within or outside the EU, for activities that are prohibited in all Member States. No funding can be granted in a Member State for an activity which is forbidden in that Member State.

The beneficiaries must pay particular attention to the principle of proportionality, the right to privacy, the right to the protection of personal data, the right to the physical and mental integrity of persons, the right to non-discrimination, the need to ensure protection of the environment and high levels of human health protection.

The beneficiaries must ensure that the activities under the action have an exclusive focus on civil applications.

The beneficiaries must ensure that the activities under the action do not:

- aim at human cloning for reproductive purposes
- intend to modify the genetic heritage of human beings which could make such modifications heritable (with the exception of research relating to cancer treatment of the gonads, which may be financed)
- intend to create human embryos solely for the purpose of research or for the purpose of stem cell procurement, including by means of somatic cell nuclear transfer, or
- lead to the destruction of human embryos (for example, for obtaining stem cells).

Activities involving research on human embryos or human embryonic stem cells may be carried out only if:

- they are set out in Annex 1 or
- the coordinator has obtained explicit approval (in writing) from the granting authority.

In addition, the beneficiaries must respect the fundamental principle of research integrity — as set out in the European Code of Conduct for Research Integrity[2].

This implies compliance with the following principles:

- reliability in ensuring the quality of research reflected in the design, the methodology, the analysis and the use of resources
- honesty in developing, undertaking, reviewing, reporting and communicating research in a transparent, fair and unbiased way

[2] European Code of Conduct for Research Integrity of ALLEA (All European Academies).

- respect for colleagues, research participants, society, ecosystems, cultural heritage and the environment
- accountability for the research from idea to publication, for its management and organisation, for training, supervision and mentoring, and for its wider impacts

and means that beneficiaries must ensure that persons carrying out research tasks follow the good research practices including ensuring, where possible, openness, reproducibility and traceability and refrain from the research integrity violations described in the Code.

Activities raising ethical issues must comply with the additional requirements formulated by the ethics panels (including after checks, reviews or audits; see Article 25).

Before starting an action task raising ethical issues, the beneficiaries must have obtained all approvals or other mandatory documents needed for implementing the task, notably from any (national or local) ethics committee or other bodies such as data protection authorities.

The documents must be kept on file and be submitted upon request by the coordinator to the granting authority. If they are not in English, they must be submitted together with an English summary, which shows that the documents cover the action tasks in question and includes the conclusions of the committee or authority concerned (if any).

VALUES (— ARTICLE 14)

Gender mainstreaming

The beneficiaries must take all measures to promote equal opportunities between men and women in the implementation of the action and, where applicable, in line with the gender equality plan. They must aim, to the extent possible, for a gender balance at all levels of personnel assigned to the action, including at supervisory and managerial level.

INTELLECTUAL PROPERTY RIGHTS (IPR) — BACKGROUND AND RESULTS — ACCESS RIGHTS AND RIGHTS OF USE (— ARTICLE 16)

Definitions

Access rights — Rights to use results or background.

Dissemination — The public disclosure of the results by appropriate means, other than resulting from protecting or exploiting the results, including by scientific publications in any medium.

Exploit(ation) — The use of results in further research and innovation activities other than those covered by the action concerned, including among other things, commercial exploitation such as developing, creating, manufacturing and marketing a product or process, creating and providing a service, or in standardisation activities.

Fair and reasonable conditions — Appropriate conditions, including possible financial terms or royalty-free conditions, taking into account the specific circumstances of the request for access, for example the actual or potential value of the results or background to which access is requested and/or the scope, duration or other characteristics of the exploitation envisaged.

FAIR principles — ‘findability’, ‘accessibility’, ‘interoperability’ and ‘reusability’.

Open access — Online access to research outputs provided free of charge to the end-user.

Open science — An approach to the scientific process based on open cooperative work, tools and diffusing knowledge.

Research data management — The process within the research lifecycle that includes the organisation, storage, preservation, security, quality assurance, allocation of persistent identifiers (PIDs) and rules and procedures for sharing of data including licensing.

Research outputs — Results to which access can be given in the form of scientific publications, data or other engineered results and processes such as software, algorithms, protocols, models, workflows and electronic notebooks.

Scope of the obligations

For this section, references to ‘beneficiary’ or ‘beneficiaries’ do not include affiliated entities (if any).

Agreement on background

The beneficiaries must identify in a written agreement the background as needed for implementing the action or for exploiting its results.

Where the call conditions restrict control due to strategic interests reasons, background that is subject to control or other restrictions by a country (or entity from a country) which is not one of the eligible countries or target countries set out in the call conditions and that impact the exploitation of the results (i.e. would make the exploitation of the results subject to control or restrictions) must not be used and must be explicitly excluded from it in the agreement on background — unless otherwise agreed with the granting authority.

Ownership of results

Results are owned by the beneficiaries that generate them.

However, two or more beneficiaries own results jointly if:

- they have jointly generated them and
- it is not possible to:
  - establish the respective contribution of each beneficiary, or
  - separate them for the purpose of applying for, obtaining or maintaining their protection.

The joint owners must agree — in writing — on the allocation and terms of exercise of their joint ownership (‘**joint ownership agreement**’), to ensure compliance with their obligations under this Agreement.

Unless otherwise agreed in the joint ownership agreement or consortium agreement, each joint owner may grant non-exclusive licences to third parties to exploit the jointly-owned results (without any right to sub-license), if the other joint owners are given:

- at least 45 days advance notice and
- fair and reasonable compensation.

The joint owners may agree — in writing — to apply another regime than joint ownership.

If third parties (including employees and other personnel) may claim rights to the results, the beneficiary concerned must ensure that those rights can be exercised in a manner compatible with its obligations under the Agreement.

The beneficiaries must indicate the owner(s) of the results (results ownership list) in the final periodic report.

Protection of results

Beneficiaries which have received funding under the grant must adequately protect their results — for an appropriate period and with appropriate territorial coverage — if protection is possible and justified, taking into account all relevant considerations, including the prospects for commercial exploitation, the legitimate interests of the other beneficiaries and any other legitimate interests.

Exploitation of results

Beneficiaries which have received funding under the grant must — up to four years after the end of the action (see Data Sheet, Point 1) — use their best efforts to exploit their results directly or to have them exploited indirectly by another entity, in particular through transfer or licensing.

If, despite a beneficiary's best efforts, the results are not exploited within one year after the end of the action, the beneficiaries must (unless otherwise agreed in writing with the granting authority) use the Horizon Results Platform to find interested parties to exploit the results.

If results are incorporated in a standard, the beneficiaries must (unless otherwise agreed with the granting authority or unless it is impossible) ask the standardisation body to include the funding statement (see Article 17) in (information related to) the standard.

Additional exploitation obligations

Where the call conditions impose additional exploitation obligations (including obligations linked to the restriction of participation or control due to strategic assets, interests, autonomy or security reasons), the beneficiaries must comply with them — up to four years after the end of the action (see Data Sheet, Point 1).

Where the call conditions impose additional exploitation obligations in case of a public emergency, the beneficiaries must (if requested by the granting authority) grant for a limited period of time specified in the request, non-exclusive licences — under fair and reasonable conditions — to their results to legal entities that need the results to address the public emergency and commit to rapidly and broadly exploit the resulting products and services at fair and reasonable conditions. This provision applies up to four years after the end of the action (see Data Sheet, Point 1).

Additional information obligation relating to standards

Where the call conditions impose additional information obligations relating to possible standardisation, the beneficiaries must — up to four years after the end of the action (see Data Sheet, Point 1) — inform the granting authority, if the results could reasonably be expected to contribute to European or international standards.

Transfer and licensing of results

Transfer of ownership

The beneficiaries may transfer ownership of their results, provided this does not affect compliance with their obligations under the Agreement.

The beneficiaries must ensure that their obligations under the Agreement regarding their results are passed on to the new owner and that this new owner has the obligation to pass them on in any subsequent transfer.

Moreover, they must inform the other beneficiaries with access rights of the transfer at least 45 days in advance (or less if agreed in writing), unless agreed otherwise in writing for specifically identified third parties including affiliated entities or unless impossible under the applicable law. This notification must include sufficient information on the new owner to enable the beneficiaries concerned to assess the effects on their access rights. The beneficiaries may object within 30 days of receiving notification (or less if agreed in writing), if they can show that the transfer would adversely affect their access rights. In this case, the transfer may not take place until agreement has been reached between the beneficiaries concerned.

Granting licences

The beneficiaries may grant licences to their results (or otherwise give the right to exploit them), including on an exclusive basis, provided this does not affect compliance with their obligations.

Exclusive licences for results may be granted only if all the other beneficiaries concerned have waived their access rights.

Granting authority right to object to transfers or licensing — Horizon Europe actions

Where the call conditions in Horizon Europe actions provide for the right to object to transfers or licensing, the granting authority may — up to four years after the end of the action (see Data Sheet, Point 1) — object to a transfer of ownership or the exclusive licensing of results, if:

- the beneficiaries which generated the results have received funding under the grant
- it is to a legal entity established in a non-EU country not associated with Horizon Europe, and
- the granting authority considers that the transfer or licence is not in line with EU interests.

Beneficiaries that intend to transfer ownership or grant an exclusive licence must formally notify the granting authority before the intended transfer or licensing takes place and:

- identify the specific results concerned
- describe in detail the new owner or licensee and the planned or potential exploitation of the results, and
- include a reasoned assessment of the likely impact of the transfer or licence on EU interests, in particular regarding competitiveness as well as consistency with ethical principles and security considerations.

The granting authority may request additional information.

If the granting authority decides to object to a transfer or exclusive licence, it must formally notify the beneficiary concerned within 60 days of receiving notification (or any additional information it has requested).

No transfer or licensing may take place in the following cases:

- pending the granting authority decision, within the period set out above
- if the granting authority objects
- until the conditions are complied with, if the granting authority objection comes with conditions.

A beneficiary may formally notify a request to waive the right to object regarding intended transfers or grants to a specifically identified third party, if measures safeguarding EU interests are in place. If the granting authority agrees, it will formally notify the beneficiary concerned within 60 days of receiving notification (or any additional information requested).

Granting authority right to object to transfers or licensing — Euratom actions

Where the call conditions in Euratom actions provide for the right to object to transfers or licensing, the granting authority may — up to four years after the end of the action (see Data Sheet, Point 1) — object to a transfer of ownership or the exclusive or non-exclusive licensing of results, if:

- the beneficiaries which generated the results have received funding under the grant
- it is to a legal entity established in a non-EU country not associated to the Euratom Research and Training Programme 2021-2025 and
- the granting authority considers that the transfer or licence is not in line with the EU interests.

Beneficiaries that intend to transfer ownership or grant a licence must formally notify the granting authority before the intended transfer or licensing takes place and:

- identify the specific results concerned
- describe in detail the results, the new owner or licensee and the planned or potential exploitation of the results, and
- include a reasoned assessment of the likely impact of the transfer or licence on EU interests, in particular regarding competitiveness as well as consistency with

ethical principles and security considerations (including the defence interests of the EU Member States under Article 24 of the Euratom Treaty).

The granting authority may request additional information.

If the granting authority decides to object to a transfer or licence, it will formally notify the beneficiary concerned within 60 days of receiving notification (or any additional information requested).

No transfer or licensing may take place in the following cases:

- pending the granting authority decision, within the period set out above
- if the granting authority objects
- until the conditions are complied with, if the granting authority objection comes with conditions.

A beneficiary may formally notify a request to waive the right to object regarding intended transfers or grants to a specifically identified third party, if measures safeguarding EU interests are in place. If the granting authority agrees, it will formally notify the beneficiary concerned within 60 days of receiving notification (or any additional information requested).

Limitations to transfers and licensing due to strategic assets, interests, autonomy or security reasons of the EU and its Member States

Where the call conditions restrict participation or control due to strategic assets, interests, autonomy or security reasons, the beneficiaries may not transfer ownership of their results or grant licences to third parties which are established in countries which are not eligible countries or target countries set out in the call conditions (or, if applicable, are controlled by such countries or entities from such countries) — unless they have requested and received prior approval by the granting authority.

The request must:

- identify the specific results concerned
- describe in detail the new owner and the planned or potential exploitation of the results, and
- include a reasoned assessment of the likely impact of the transfer or license on the strategic assets, interests, autonomy or security of the EU and its Member States.

The granting authority may request additional information.

Access rights to results and background

Exercise of access rights — Waiving of access rights — No sub-licensing

Requests to exercise access rights and the waiver of access rights must be in writing.

Unless agreed otherwise in writing with the beneficiary granting access, access rights do not include the right to sub-license.

If a beneficiary is no longer involved in the action, this does not affect its obligations to grant access.

If a beneficiary defaults on its obligations, the beneficiaries may agree that that beneficiary no longer has access rights.

Access rights for implementing the action

The beneficiaries must grant each other access — on a royalty-free basis — to background needed to implement their own tasks under the action, unless the beneficiary that holds the background has — before acceding to the Agreement —:

- informed the other beneficiaries that access to its background is subject to restrictions, or
- agreed with the other beneficiaries that access would not be on a royalty-free basis.

The beneficiaries must grant each other access — on a royalty-free basis — to results needed for implementing their own tasks under the action.

Access rights for exploiting the results

The beneficiaries must grant each other access — under fair and reasonable conditions — to results needed for exploiting their results.

The beneficiaries must grant each other access — under fair and reasonable conditions — to background needed for exploiting their results, unless the beneficiary that holds the background has — before acceding to the Agreement — informed the other beneficiaries that access to its background is subject to restrictions.

Requests for access must be made — unless agreed otherwise in writing — up to one year after the end of the action (see Data Sheet, Point 1).

Access rights for entities under the same control

Unless agreed otherwise in writing by the beneficiaries, access to results and, subject to the restrictions referred to above (if any), background must also be granted — under fair and reasonable conditions — to entities that:

- are established in an EU Member State or Horizon Europe associated country
- are under the direct or indirect control of another beneficiary, or under the same direct or indirect control as that beneficiary, or directly or indirectly controlling that beneficiary and
- need the access to exploit the results of that beneficiary.

Unless agreed otherwise in writing, such requests for access must be made by the entity directly to the beneficiary concerned.

Requests for access must be made — unless agreed otherwise in writing — up to one year after the end of the action (see Data Sheet, Point 1).

Access rights for the granting authority, EU institutions, bodies, offices or agencies and national authorities to results for policy purposes — Horizon Europe actions

In Horizon Europe actions, the beneficiaries which have received funding under the grant must grant access to their results — on a royalty-free basis — to the granting authority, EU institutions, bodies, offices or agencies for developing, implementing and monitoring EU policies or programmes. Such access rights do not extend to beneficiaries' background.

Such access rights are limited to non-commercial and non-competitive use.

For actions under the cluster 'Civil Security for Society', such access rights also extend to national authorities of EU Member States for developing, implementing and monitoring their policies or programmes in this area. In this case, access is subject to a bilateral agreement to define specific conditions ensuring that:

- the access rights will be used only for the intended purpose and
- appropriate confidentiality obligations are in place.

Moreover, the requesting national authority or EU institution, body, office or agency (including the granting authority) must inform all other national authorities of such a request.

Access rights for the granting authority, Euratom institutions, funding bodies or the Joint Undertaking Fusion for Energy — Euratom actions

In Euratom actions, the beneficiaries which have received funding under the grant must grant access to their results — on a royalty-free basis — to the granting authority, Euratom institutions, funding bodies or the Joint Undertaking Fusion for Energy for developing, implementing and monitoring Euratom policies and programmes or for compliance with obligations assumed through international cooperation with non-EU countries and international organisations.

Such access rights include the right to authorise third parties to use the results in public procurement and the right to sub-license and are limited to non-commercial and non-competitive use.

Additional access rights

Where the call conditions impose additional access rights, the beneficiaries must comply with them.

COMMUNICATION, DISSEMINATION, OPEN SCIENCE AND VISIBILITY (— ARTICLE 17)

Dissemination

Dissemination of results

The beneficiaries must disseminate their results as soon as feasible, in a publicly available format, subject to any restrictions due to the protection of intellectual property, security rules or legitimate interests.

A beneficiary that intends to disseminate its results must give at least 15 days advance notice to the other beneficiaries (unless agreed otherwise), together with sufficient information on the results it will disseminate.

Any other beneficiary may object within (unless agreed otherwise) 15 days of receiving notification, if it can show that its legitimate interests in relation to the results or background would be significantly harmed. In such cases, the results may not be disseminated unless appropriate steps are taken to safeguard those interests.

Additional dissemination obligations

Where the call conditions impose additional dissemination obligations, the beneficiaries must also comply with those.

Open Science

Open science: open access to scientific publications

The beneficiaries must ensure open access to peer-reviewed scientific publications relating to their results. In particular, they must ensure that:

- at the latest at the time of publication, a machine-readable electronic copy of the published version or the final peer-reviewed manuscript accepted for publication, is deposited in a trusted repository for scientific publications

- immediate open access is provided to the deposited publication via the repository, under the latest available version of the Creative Commons Attribution International Public Licence (CC BY) or a licence with equivalent rights; for monographs and other long-text formats, the licence may exclude commercial uses and derivative works (e.g. CC BY-NC, CC BY-ND) and

- information is given via the repository about any research output or any other tools and instruments needed to validate the conclusions of the scientific publication.

Beneficiaries (or authors) must retain sufficient intellectual property rights to comply with the open access requirements.

Metadata of deposited publications must be open under a Creative Common Public Domain Dedication (CC 0) or equivalent, in line with the FAIR principles (in particular machine-actionable) and provide information at least about the following: publication (author(s), title, date of publication, publication venue); Horizon Europe or Euratom funding; grant project name, acronym and number; licensing terms; persistent identifiers for the publication, the authors involved in the action and, if possible, for their organisations and the grant. Where applicable, the metadata must include persistent identifiers for any research output or any other tools and instruments needed to validate the conclusions of the publication.

Only publication fees in full open access venues for peer-reviewed scientific publications are eligible for reimbursement.

Open science: research data management

The beneficiaries must manage the digital research data generated in the action ('data') responsibly, in line with the FAIR principles and by taking all of the following actions:

- establish a data management plan ('DMP') (and regularly update it)

- as soon as possible and within the deadlines set out in the DMP, deposit the data in a trusted repository; if required in the call conditions, this repository must be federated in the EOSC in compliance with EOSC requirements

- as soon as possible and within the deadlines set out in the DMP, ensure open access — via the repository — to the deposited data, under the latest available version of the Creative Commons Attribution International Public License (CC BY) or Creative Commons Public Domain Dedication (CC 0) or a licence with equivalent rights, following the principle 'as open as possible as closed as necessary', unless providing open access would in particular:

    - be against the beneficiary's legitimate interests, including regarding commercial exploitation, or

    - be contrary to any other constraints, in particular the EU competitive interests or the beneficiary's obligations under this Agreement; if open access is not provided (to some or all data), this must be justified in the DMP

- provide information via the repository about any research output or any other tools and instruments needed to re-use or validate the data.

Metadata of deposited data must be open under a Creative Common Public Domain Dedication (CC 0) or equivalent (to the extent legitimate interests or constraints are safeguarded), in line with the FAIR principles (in particular machine-actionable) and provide information at least about the following: datasets (description, date of deposit, author(s), venue and embargo); Horizon Europe or Euratom funding; grant project name, acronym and number; licensing terms; persistent identifiers for the dataset, the authors involved in the action, and, if possible, for their organisations and the grant. Where applicable, the metadata must include persistent identifiers for related publications and other research outputs.

Open science: additional practices

Where the call conditions impose additional obligations regarding open science practices, the beneficiaries must also comply with those.

Where the call conditions impose additional obligations regarding the validation of scientific publications, the beneficiaries must provide (digital or physical) access to data or other results needed for validation of the conclusions of scientific publications, to the extent that their legitimate interests or constraints are safeguarded (and unless they already provided the (open) access at publication).

Where the call conditions impose additional open science obligations in case of a public emergency, the beneficiaries must (if requested by the granting authority) immediately deposit any research output in a repository and provide open access to it under a CC BY licence, a Public Domain Dedication (CC 0) or equivalent. As an exception, if the access would be against the beneficiaries' legitimate interests, the beneficiaries must grant non-exclusive licenses — under fair and reasonable conditions — to legal entities that need the research output to address the public emergency and commit to rapidly and broadly exploit the resulting products and services at fair and reasonable conditions. This provision applies up to four years after the end of the action (see Data Sheet, Point 1).

Plan for the exploitation and dissemination of results including communication activities

Unless excluded by the call conditions, the beneficiaries must provide and regularly update a plan for the exploitation and dissemination of results including communication activities.

SPECIFIC RULES FOR CARRYING OUT THE ACTION (— ARTICLE 18)

Implementation in case of restrictions due to strategic assets, interests, autonomy or security of the EU and its Member States

Where the call conditions restrict participation or control due to strategic assets, interests, autonomy or security, the beneficiaries must ensure that none of the entities that participate as affiliated entities, associated partners, subcontractors or recipients of financial support to third parties are established in countries which are not eligible countries or target countries set out in the call conditions (or, if applicable, are controlled by such countries or entities from such countries) — unless otherwise agreed with the granting authority.

The beneficiaries must moreover ensure that any cooperation with entities established in countries which are not eligible countries or target countries set out in the call conditions (or, if applicable, are controlled by such countries or entities from such countries) does not affect the strategic assets, interests, autonomy or security of the EU and its Member States.

Recruitment and working conditions for researchers

The beneficiaries must take all measures to implement the principles set out in the Commission Recommendation on the European Charter for Researchers and the Code of Conduct for the Recruitment of Researchers[3], in particular regarding:

- working conditions
- transparent recruitment processes based on merit, and
- career development.

The beneficiaries must ensure that researchers and all participants involved in the action are aware of them.

Specific rules for access to research infrastructure activities

Definitions

Research Infrastructures — Facilities that provide resources and services for the research communities to conduct research and foster innovation in their fields. This definition includes the associated human resources, and it covers major equipment or sets of instruments; knowledge-related facilities such as collections, archives or scientific data infrastructures; computing systems, communication networks, and any other infrastructure, of a unique nature and open to external users, essential to achieve excellence in research and innovation. Where relevant, they may be used beyond research, for example

[3] Commission Recommendation 2005/251/EC of 11 March 2005 on the European Charter for Researchers and on a Code of Conduct for the Recruitment of Researchers (OJ L 75, 22.3.2005, p. 67).

for education or public services, and they may be 'single-sited', 'virtual' or 'distributed'[4]:

When implementing access to research infrastructure activities, the beneficiaries must respect the following conditions:

- for transnational access:

  - access which must be provided:

    The access must be free of charge, transnational access to research infrastructure or installations for selected user-groups.

    The access must include the logistical, technological and scientific support and the specific training that is usually provided to external researchers using the infrastructure. Transnational access can be either in person (hands-on), provided to selected users that visit the installation to make use of it, or remote, through the provision to selected user-groups of remote scientific services (e.g. provision of reference materials or samples, remote access to a high-performance computing facility).

  - categories of users that may have access:

    Transnational access must be provided to selected user-groups, i.e. teams of one or more researchers (users).

    The majority of the users must work in a country other than the country(ies) where the installation is located (unless access is provided by an international organisation, the Joint Research Centre (JRC), an ERIC or similar legal entity).

    Only user groups that are allowed to disseminate the results they have generated under the action may benefit from the access (unless the users are working for SMEs).

    Access for user groups with a majority of users not working in a EU Member State or Horizon Europe associated country is limited to 20% of the total amount of units of access provided under the grant (unless a higher percentage is foreseen in Annex 1).

  - procedure and criteria for selecting user groups:

    The user groups must request access by submitting (in writing) a description of the work that they wish to carry out and the names, nationalities and home institutions of the users.

    The user groups must be selected by (one or more) selection panels set up by the consortium.

[4] See Article 2(1) of the Horizon Europe Framework Programme Regulation 2021/695.

The selection panels must be composed of international experts in the field, at least half of them independent from the consortium (unless otherwise specified in Annex 1).

The selection panels must assess all proposals received and recommend a short-list of the user groups that should benefit from access.

The selection panels must base their selection on scientific merit, taking into account that priority should be given to user groups composed of users who:

- have not previously used the installation and
- are working in countries where no equivalent research infrastructure exist.

It will apply the principles of transparency, fairness and impartiality.

Where the call conditions impose additional rules for the selection of user groups, the beneficiaries must also comply with those.

- other conditions:

The beneficiaries must request written approval from the granting authority for the selection of user groups requiring visits to the installations exceeding 3 months (unless such visits are foreseen in Annex 1).

In addition, the beneficiaries must:

- advertise widely, including on a their websites, the access offered under the Agreement
- promote equal opportunities in advertising the access and take into account the gender dimension when defining the support provided to users
- ensure that users comply with the terms and conditions of the Agreement
- ensure that its obligations under Articles 12, 13, 17 and 33 also apply to the users
- keep records of the names, nationalities, and home institutions of users, as well as the nature and quantity of access provided to them

- for virtual access:

  - access which must be provided:

The access must be free of charge, virtual access to research infrastructure or installations.

‘Virtual access’ means open and free access through communication networks to digital resources and services needed for research, without selecting the users to whom access is provided.

The access must include the support that is usually provided to external users.

Where allowed by the call conditions, beneficiaries may in justified cases define objective eligibility criteria (e.g. affiliation to a research or academic institution) for specific users.

- other conditions:

  The beneficiaries must have the virtual access services assessed periodically by a board composed of international experts in the field, at least half of whom must be independent from the consortium (unless otherwise specified in Annex 1). For this purpose, information and statistics on the users and the nature and quantity of the access provided, must be made available to the board.

  The beneficiaries must advertise widely, including on a dedicated website, the access offered under the grant and the eligibility criteria, if any.

  Where the call conditions impose additional traceability[5] obligations, information on the traceability of the users and the nature and quantity of access must be provided by the beneficiaries.

These obligations apply regardless of the form of funding or budget categories used to declare the costs (unit costs or actual costs or a combination of the two).

[5] According to the definition given in ISO 9000, i.e.: "Traceability is the ability to trace the history, application, use and location of an item or its characteristics through recorded identification data." The users can be traced, for example, by authentication and/or by authorization or by other means that allows for analysis of the type of users and the nature and quantity of access provided.

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